

README

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Z80 ROMless SBC - WIP Engineering & Build Specification

Repository: github.com/gloveboxes/Z80ROMlessSBC PDF edition: [README.pdf](#)

Overview

This document describes a theoretical Z80 single-board computer design that has not yet been built, wired, or validated in hardware. Treat it as an engineering proposal and bring-up plan, not as a proven reference design. Every electrical assumption, timing margin, and firmware interaction still needs bench validation with the staged tests in Section 8 before the design should be considered reliable.

The proposed system is a ROMless Z80 computer supervised by a Raspberry Pi Pico 2 W. The Pico supplies the Z80 clock, holds and releases reset, loads a boot image directly into SRAM by taking the Z80 bus, and then lets the Z80 execute from RAM. A 64 KB SRAM provides the Z80 memory space; level shifters and bus transceivers isolate the Pico's 3.3 V GPIO from the 5 V Z80/SRAM bus; an MCP23S17 expands the Pico's address-drive capability during DMA and trapped I/O cycles.

The key design idea is that the Pico acts as both supervisor and virtual peripheral controller. It can stop the fully static Z80 clock on an I/O cycle, inspect the requested port, exchange one data byte, and then resume execution. Terminal I/O is intended to be one of those virtual peripherals: Z80 IN and OUT instructions feed nonblocking queues on the Pico, while a WebSocket terminal server runs on the Pico's other core so Wi-Fi and browser traffic do not disturb Z80 timing.

Z80 software lives in a reserved region of the Pico's own onboard flash rather than on removable media or compiled into the running firmware image. The Pico reads boot binaries, monitor images, and CP/M disk images directly from that memory-mapped flash region, then uses the already-validated DMA path to populate SRAM before the Z80 is released. Once the Z80 is running, the same I/O trap mechanism can expose sector-oriented virtual disk ports backed by that same flash partition (Section 6.3).

The build plan is deliberately phased. Early phases prove power, translation, SPI expansion, bus isolation, SRAM DMA, and clock control before the Z80 is installed. Later phases prove virtual-ROM boot, synchronous I/O trapping, WebSocket terminal service, and finally the maximum qualified clock rate. Passing an earlier phase is a prerequisite for trusting the assumptions used by the next one.

Firmware and CP/M build

The repository implements the ten cumulative firmware stages under `src/`. Stage 10 includes the journaled flash disks, CP/M boot loader, and WebSocket terminal. Its host build also assembles the board-native CP/M 2.2 BIOS, reconstructs pristine CCP/BDOS bytes from the preserved Altair source image, and emits all provisionable images.

Install CMake, Ninja, Python 3, `picotool`, and `z80asm`. The firmware needs a complete Arm GNU bare-metal toolchain with Newlib; the Homebrew compiler alone does not provide the required runtime. A known working setup is:

```
brew install cmake ninja picotool z80asm
export PATH="$HOME/.local/share/arm-gnu-toolchain-15.3.rel1/bin:$PATH"
cmake -S . -B build -G Ninja \
  -DPICO_BOARD=pico2_w \
  -DZ80_WIFI_SSID='your-network' \
  -DZ80_WIFI_PASSWORD='your-password'
cmake --build build --target z80_cpm_images -j
```

Wi-Fi credentials are written only to the generated build tree. An empty SSID leaves networking disabled while flash-disk service continues to operate. The `z80_cpm_images` target builds Stage 10 and writes these files to `build/cpm/`:

Artifact	Purpose
z80boot.pkg	Manifest, CRCs, and the reset-ready 64 KiB Z80 image
drive_a_cpm63k.img	Native CP/M system disk with the SBC BIOS
drive_b_bdsc.img through drive_d_blank.img	Converted 320 KiB disks
z80romless-flash.bin	Complete 4 MiB initial-provisioning image
manifest.json	Geometry, addresses, sizes, and SHA-256 values

Run the host regression checks with:

```
python3 -m unittest discover -s src/cpm -p 'test_*.py' -v
```

PDF edition

The checked-in README.pdf is generated from this file. The renderer preserves tables, highlighted code, local images, links, and MathML. Each Mermaid diagram gets a dedicated page; wide diagrams use A4 landscape pages and are rendered as vector graphics without the interactive preview controls.

Install Node.js, Pandoc, and a Chromium-family browser, then run:

```
brew install node pandoc
npm ci
npm run pdf
```

The script searches for Microsoft Edge, Google Chrome, or Chromium. Set BROWSER_EXECUTABLE to use another installation. Optional input and output paths may be passed after --, for example:

```
npm run pdf -- README.md README.pdf
```

Flash provisioning

For a new or disposable flash, put the Pico in BOOTSEL mode and write the complete image. This erases existing CP/M disk contents and the journal:

```
picotool load -v build/cpm/z80romless-flash.bin -t bin -o 0x1000000
picotool reboot
```

For normal updates, load only the firmware and selected storage regions. These commands preserve regions not named by the input file:

```
picotool load -v build/src/stage10_websocket_terminal/z80_stage10_websocket_terminal.uf2
picotool load -v build/cpm/z80boot.pkg -t bin -o 0x102A0000
picotool load -v build/cpm/drive_a_cpm63k.img -t bin -o 0x102C0000
picotool load -v build/cpm/drive_b_bdsc.img -t bin -o 0x10310000
picotool load -v build/cpm/drive_c_escape.img -t bin -o 0x10360000
picotool load -v build/cpm/drive_d_blank.img -t bin -o 0x103B0000
picotool reboot
```

After association, browse to <http://<pico-dhcp-address>:8088/>. The USB serial console reports Stage 10 startup and accepts s to print terminal queue/drop counts and disk/fatal status. CP/M disk writes are persistent, so retain host backups before full reprovisioning.

0. Project Inventory

The quantities below build one complete three-breadboard prototype. Purchase quantities include a small allowance for breadboard spares.

0.1 Semiconductors

Required	Component	Package	Function
1	Raspberry Pi Pico 2 W	Module with two 20-pin headers	Supervisor, clock, DMA, virtual I/O, and Wi-Fi terminal
1	Z84C0020PEC	40-pin PDIP	CMOS Z80 CPU
1	AS6C1008-55PCN	32-pin PDIP	SRAM; lower 64 KB used
1	MCP23S17-E/SP	28-pin SPDIP	SPI-to-16-bit address interface
1	ATF22V10B-15PC or ATF22V10C-15PU	24-pin PDIP	Programmable SRAM control-source arbitration
1	SN74HCT541N	20-pin PDIP	Eight-channel 5 V output buffer for clock, BUSREQ#, SPI, and SRAM controls
1	SN74AHCT245N	20-pin PDIP	Fixed Pico-to-5 V data-bus path; A side faces Pico, B side faces bus
1	SN74LVC245AN	20-pin PDIP	Fixed 5 V-to-Pico data-bus path with 5 V-tolerant inputs and Ioff protection
1	SN74LVC244AN	20-pin PDIP	5 V-to-3.3 V buffering for Z80 status/control and MCP SO
1	2N3904	TO-92	Pulls MCP23S17 RESET# LOW whenever GAL address access is disabled
1	1N5819	Axial diode	Schottky power OR from external +5 V to Pico VSYS

Nine-package all-through-hole design: The ATF22V10 computes all three SRAM controls from RESET#, BUSACK#, the Pico DMA controls, and the Z80 MREQ#/RD#/WR# outputs (Section 1.2). The ATF22V10 guarantees only a TTL-level 2.4 V HIGH, so its SRAM outputs pass through the SN74HCT541N rather than driving the 5 V SRAM directly. The same HCT541 translates Pico CLK, BUSREQ#, and the three MCP23S17 SPI inputs to guaranteed 5 V levels. Pico GP3 drives the Z80 RESET# input directly because the Z84C00 specifies a 2.2 V HIGH threshold for non-clock inputs; the Z80 clock uses the HCT541 path to satisfy its stricter threshold. The data bus uses separate fixed-direction SN74AHCT245N and SN74LVC245AN devices because no single PDIP part provides a dual-supply, bidirectional, power-off-safe function. Two spare GAL inputs accept the existing DIR and master-enable controls, and two spare GAL outputs generate mutually exclusive active-low OEs. A third spare GAL output and one 2N3904 hold the MCP23S17 in reset outside DMA/trap access, allowing its reset-default input ports to connect directly to pulled-up A0-A15 and safely share the address bus. The design uses 9 active packages with deterministic isolation and safe power sequencing. The SN74LVC244AN buffers RD#/WR# because Pico 2 GP27 and GP28 are standard ADC-capable pads, not 5 V-tolerant FT pads. It also buffers BUSACK#, IORQ#, and MCP SO: although GP0, GP1, and GP20 are FT pads, their 5.5 V tolerance requires RP2350 IOVDD to be present at 3.3 V. Using the spare LVC244 channels preserves safe power sequencing and powered-off isolation (Section 5.3).

0.2 Sockets and Headers

Quantity	Item
1	40-pin DIP socket
1	32-pin DIP socket
1	28-pin DIP socket

Quantity	Item
1	24-pin, 0.3-inch-wide DIP socket for the ATF22V10
4	20-pin DIP sockets
2	20-pin 0.1-inch male headers for the Pico 2, if not fitted
2	20-position 0.1-inch socket strips for removable Pico mounting; do not substitute DIP IC sockets

0.3 Capacitors and Resistors

Fitted	Purchase	Item	Purpose
8	11	100 nF X7R ceramic capacitors, at least 10 V	One at every DIP logic supply pair
3	5	22 uF capacitors, at least 10 V	One per breadboard
1	2	100 uF electrolytic capacitor, at least 10 V	5 V supply-entry bulk capacitance
29	35	10 kOhm, 1/4 W resistors	Defined startup levels and temporary test pulls
1	3	4.7 kOhm, 1/4 W resistor	GAL-to-2N3904 base current limiting
1	3	47 kOhm, 1/4 W resistor	2N3904 base-emitter pull-down
3	4	8x10 kOhm bussed SIP resistor networks, 9-pin	Two A0-A15 pull-ups (common to +5 V) and one Pico D0-D7 pull-down (common to GND)
Reused for tests	10	1 kOhm, 1/4 W resistors	First-drive current limiting and manual input tests

Place each 100 nF capacitor directly across its IC supply pins with the shortest practical leads. If scope captures show more than 250 mV rail excursion at the farthest board, add 47-100 uF there and repeat the capture; bulk capacitance does not replace local 100 nF capacitors. On the 5 V side, fit 10 kOhm pull-ups to BUSREQ#, BUSACK#, MREQ#, IORQ#, RD#, WR#, MCP SO, SRAM CE#, SRAM OE#, SRAM WE#, WAIT#, INT#, and NMI#. RESET# is the direct GP3 node and must not have a 5 V pull-up. Also pull SN74HCT541 A6, A7, and A8 up to 5 V so its SRAM-control outputs remain inactive if the GAL is physically absent. Fit 10 kOhm pull-ups on A0-A15 using two 9-pin bussed SIP networks. On the Pico side, use 10 kOhm pull-ups to 3.3 V on GP4 (BUSREQ#), GP5 (SRAM CE#), GP21 (SPI CS#), GP22 (SRAM WE#), and GP26 (SRAM OE#). Use 10 kOhm pull-downs to GND on GP2 (CLK), GP3 (RESET#), GP6 (DATA_DIR), GP7 (DATA_ENABLE), GP9 (ADDR_ENABLE), and GP18/GP19 (SPI SCK/SI). Leave GP8 unconnected. Connect GP7 and GP6 directly to ATF22V10 pins 9 and 11; its TTL-compatible inputs accept 3.3 V. Fit the third 8x10 kOhm bussed SIP network from GP10-GP17 to GND so the SN74AHCT245N A inputs stay defined while Pico GPIOs are inputs or the Pico is absent. It loads each active HIGH by only 0.33 mA at 3.3 V. Once the Pico 3.3 V rail is valid, these external resistors establish safe levels before the firmware configures SIO. During a cold power ramp the Pico-side pull-ups cannot hold active-low controls HIGH while their 3.3 V rail is still at 0 V; RESET# remains asserted and SRAM contents are therefore treated as indeterminate until the boot image is loaded and verified. The HCT541-input pull-ups cover an absent GAL, and the final SRAM-node pull-ups cover an absent HCT541. They do not provide power-off isolation for an installed source IC. Tie every unused CMOS input to a defined level.

0.4 Construction and Power

Quantity	Item	Requirement
3	830-tie-point solderless breadboards (BusBoard/X-ON BB830, ABS body; 63 terminal rows plus two 100-point power-distribution strips per board)	Memory, CPU core, and peripheral zones
1	Regulated 5 V supply	At least 1 A; adjustable current limiting preferred
1	USB data cable	Pico programming and serial diagnostics
1 set	22 AWG solid-core hookup wire	Multiple colors for signal groups
1 set	Short male-to-male jumpers	Inter-board and temporary connections
1 set	Short ground and power-distribution links	Connects all three boards; includes +5 V, 3.3 V, and multiple GND links

Quantity	Item	Requirement
1	In-line power switch or removable supply jumper	Emergency power removal; rated at least 1.5 A continuous
As required	Labels or wire markers	Bus bits, controls, and socket pin 1

Feed the breadboard's +5 V logic rail directly from the regulated supply. Feed Pico VSYS from that rail only through the 1N5819: anode to external +5 V, banded cathode to VSYS. Pico 2 already has a Schottky diode from USB VBUS to VSYS, so this second diode safely ORs USB and external power without back-powering either source. Never link the external +5 V rail directly to Pico VBUS or VSYS. Feed the SN74LVC245AN and SN74LVC244AN from the Pico 3.3 V rail. Connect the SN74AHCT245N and all other logic to the regulated 5 V rail. No 5 V output connects directly to a Pico GPIO; both LVC devices provide power-off isolation. The GAL inputs do not source 5 V into GP6/GP7, and the SN74LVC244AN's I_{off} protection keeps the five monitored inputs isolated while its 3.3 V supply is absent or ramping. Tie the Pico's AGND pin (pin 33) to the common digital ground: this design does not use the ADC function on GP26-GP29, so no separate analogue ground plane is needed.

All installed 5 V logic devices must be powered from the same 5 V rail whenever that rail is energized. In particular, do not apply 5 V with an installed ATF22V10 or SN74HCT541 unpowered: downstream pull-ups can otherwise raise an output above its $V_{CC} + 0.75\text{ V}$ GAL limit or the corresponding logic-family absolute maximum. The absent-device pull-up behavior is valid only when that device is physically removed from its socket. Verify 5 V continuity at every installed IC before power-up; a missing VCC socket contact is a fault, not a supported operating mode.

The plug-in supply shown on the middle breadboard may provide the 5 V entry point only after a load test proves 4.75 V to 5.25 V at the board under at least 500 mA load without regulator overheating. Do not connect that module's 3.3 V output to the Pico 3V3 pin or the system 3.3 V rail; the Pico must remain the only 3.3 V source. Reserve terminal rows 1-3 under the module body even if its pins enter only the distribution strip. If the actual module obscures more than three terminal rows, move it off-board or to the Memory Board rather than compressing the Core Board placement.

0.5 Bring-Up Equipment

- Digital multimeter with resistance, continuity, and DC voltage modes.
- Two-channel oscilloscope rated for at least 100 MHz for edge-integrity checks; a 20 MHz instrument is adequate only for low-speed functional bring-up.
- Logic analyzer with at least 16 channels; 24 or more is preferred.
- Current-limited bench supply or an in-line 5 V current meter.
- Fine probe hooks or test clips suitable for DIP pins.
- A programmer explicitly supporting the selected ATF22V10B or ATF22V10C device and generic 22V10 JEDEC files.

0.6 Optional Items

- LEDs with 1 kOhm series resistors for low-speed diagnostics only; do not permanently load high-speed buses with LEDs.
- No card socket or extra storage hardware is needed for boot images or CP/M disks: they live in a reserved region of the Pico's own onboard flash (Section 6.3), provisioned once with `picotool` and read back over the existing memory-mapped XIP bus. Reflash that region with `picotool` to change a disk image; no wiring changes.
- Spare 100 nF capacitors, resistors, jumper wire, and an IC extractor.

1. Reference Pin Mapping & Logic Domain Verification

This architecture leverages the fully static nature of the Z84C0020PEC CMOS CPU. The Pico controls the clock and uses the Z80 BUSREQ#/ BUSACK# handshake for DMA ownership. During RESET, the address and data buses float and control outputs are inactive. During BUSACK#, A0-A15, D0-D7, MREQ#, IORQ#, RD#, and WR# float while BUSACK# remains actively driven LOW; M1#, RFSH#, and HALT# are not part of the floated bus set.

1.0 Raspberry Pi Pico 2 Header Pin Map

GPIO numbers in the firmware are not physical header numbers. Use this table for construction; the ranges preserve ascending bit order.

Pico function	GPIO	Physical header pin	Connection
BUSACK# monitor	GP0	1	SN74LVC244AN 1Y1
IORQ# trap input	GP1	2	SN74LVC244AN 1Y2
Z80 CLK	GP2	4	SN74HCT541 A1 pin 2
Z80 RESET#	GP3	5	Z80 pin 26 and ATF22V10 pin 1, direct 3.3 V logic
Z80 BUSREQ#	GP4	6	SN74HCT541 A2 pin 3
SRAM CE#	GP5	7	ATF22V10 pin 7
Data DIR / ENABLE	GP6 / GP7	9 / 10	ATF22V10 pins 11 / 9; GAL pins 17 / 18 generate mutually exclusive data OE# signals
Unused	GP8	11	Leave open; no address transceivers are fitted
Address enable	GP9	12	ATF22V10 pin 10; HIGH releases MCP RESET# through GAL pin 19 and Q1
D0-D3	GP10- GP13	14-17	SN74AHCT245N and SN74LVC245AN A1-A4 pins 2-5
D4-D7	GP14- GP17	19-22	SN74AHCT245N and SN74LVC245AN A5-A8 pins 6-9
SPI SCK / MOSI / MISO / CS#	GP18- GP21	24-27	HCT541 A4/A5, LVC244 2Y1, HCT541 A3
SRAM WE#	GP22	29	ATF22V10 pin 3
SRAM OE#	GP26	31	ATF22V10 pin 5
Z80 RD# monitor	GP27	32	SN74LVC244AN 1Y3
Z80 WR# monitor	GP28	34	SN74LVC244AN 1Y4
3.3 V output	-	36	SN74LVC245AN VCC, LVC244 VCC, and 3.3 V pull-ups
VSYS	-	39	1N5819 banded cathode; anode to external +5 V

Connect Pico GND pins 3, 8, 13, 18, 23, 28, and 38, plus AGND pin 33, to common ground with multiple short links. Leave RUN pin 30, ADC_VREF pin 35, and 3V3_EN pin 37 open. Do not connect the external 5 V rail to VBUS pin 40; USB supplies that node internally.

Component	Signal Name / Group	Hardware Pin Numbers	Electrical Constraint / Logic Domain
Z84C0020PEC (CPU)	CLK	Pin 6	Input. Driven by level-shifted 5V CMOS square wave. Clock can be safely frozen indefinitely in either a HIGH or LOW state.
	IORQ#	Pin 20	Output. Active Low. Buffered to 3.3 V through SN74LVC244AN channel 1A2/1Y2 and monitored by Pico 2 GP1 to trigger clock-stop trapping.
	MREQ#	Pin 19	Output. Active Low, three-state. Connects to ATF22V10 pin 8 for CPU-owned SRAM cycles; pulled HIGH when the CPU is absent.
	RD#	Pin 21	Output. Active Low. Buffered to 3.3 V through SN74LVC244AN channel 1A3/1Y3 and sampled by Pico 2 GP27 to resolve cycle intent.
	WR#	Pin 22	Output. Active Low. Buffered to 3.3 V through SN74LVC244AN channel 1A4/1Y4 and sampled by Pico 2 GP28 to resolve cycle intent alongside RD#.
	BUSACK#	Pin 23	Output. Active Low. Feeds ATF22V10 pin 2 for SRAM-source arbitration (Section 1.2) and is buffered to 3.3 V through SN74LVC244AN channel 1A1/1Y1 for monitoring at Pico 2 GP0.
	BUSREQ#	Pin 25	Input. Active Low. Level-shifted up to 5V to request DMA ownership.
	RESET#	Pin 26	Input. Active Low. Driven directly by Pico GP3 at TTL-compatible 3.3 V and also connected to ATF22V10 pin 1 for SRAM-source arbitration. Hold LOW for at least three full clock cycles.
	M1#	Pin 27	Output. Active Low. Probe point for the Phase 7 opcode-fetch test; not connected to Pico logic.
	A0 – A15 (Address Bus)	Pins 30 – 40, 1 – 5	5V Tri-state Bus. Driven by CPU during active run, floats during DMA/RESET.
	D0 – D7 (Data Bus)	Pins 7 – 10, 12 – 15	5V Bi-directional Tri-state Bus. Connected directly to SRAM data bus pins.
	WAIT#	Pin 24	Input. Active Low. Unused; pull to +5V through 10 kOhm so it never floats LOW and stalls the CPU in a permanent wait state.
	INT#	Pin 16	Input. Active Low. Unused; pull to +5V through 10 kOhm to prevent a spurious interrupt from a floating input.
	NMI#	Pin 17	Input. Active Low, edge-sensed. Unused; pull to +5V through 10 kOhm to prevent a spurious non-maskable interrupt from a floating input.

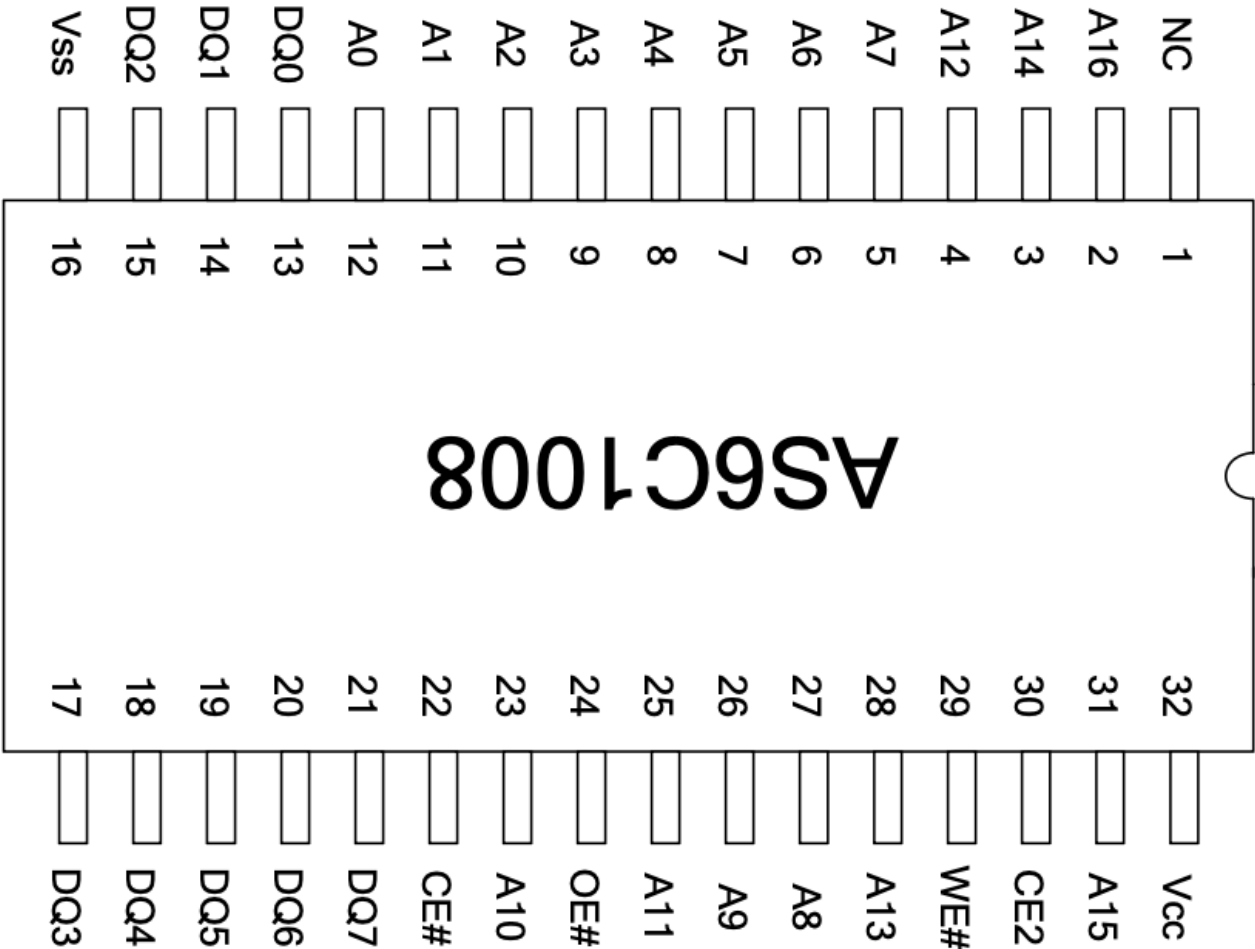
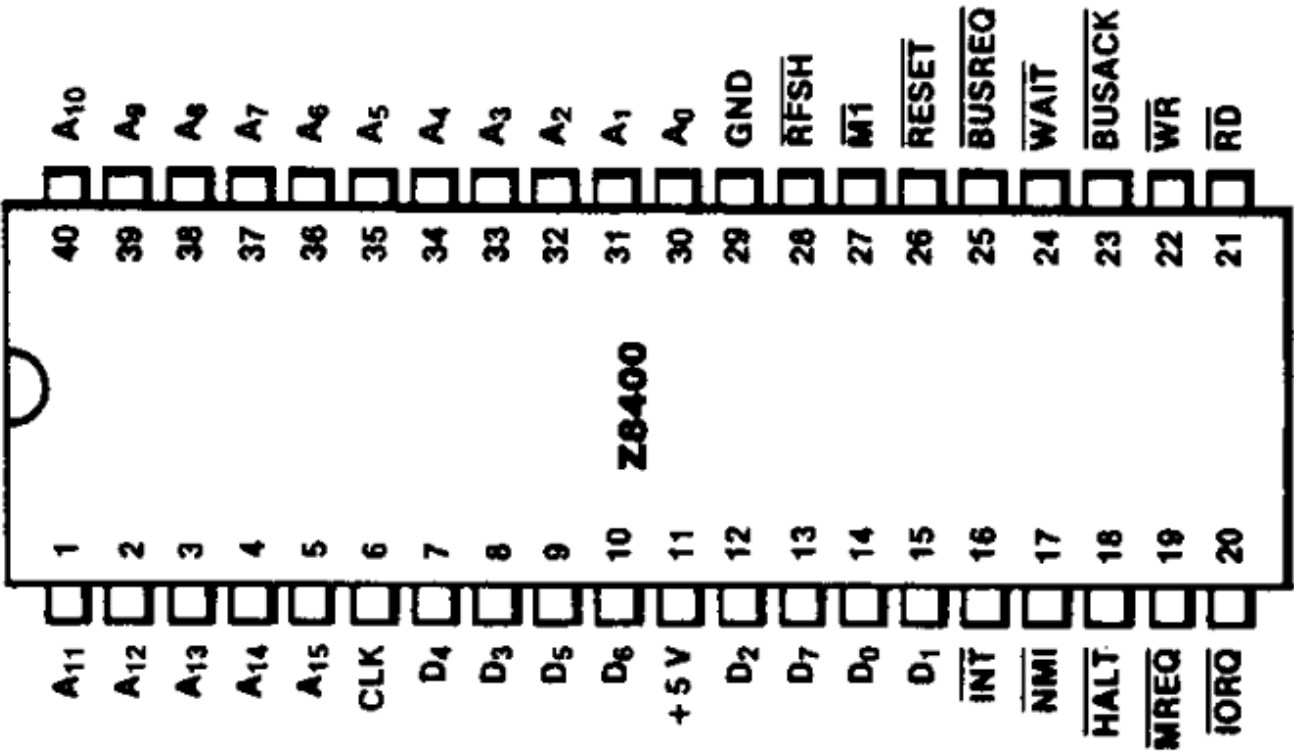
Component	Signal Name / Group	Hardware Pin Numbers	Electrical Constraint / Logic Domain
AS6C1008-55PCN (SRAM)	A0 – A15 (Address lines)	Pins 12-10, 9-7, 6-5, 27-26, 23, 25, 4, 28, 3, 31	5V Input. Connected directly to the 5V Z80 address bus. Driven by MCP23S17 during DMA.
	A16	Pin 2	5V Input. Tie to GND to select the lower 64KB bank; do not leave floating.
	D0 – D7 (Data lines)	Pins 13 – 15, 17 – 21	5V Bi-directional Bus. Connected directly to the Z80 data bus.
	WE#	Pin 29	Input. Active Low. Driven by SN74HCT541 Y6 from the ATF22V10 arbitration output (Section 1.2): Z80 WR# during CPU ownership, Pico DMA write signal during DMA ownership.
	OE#	Pin 24	Input. Active Low. Driven by SN74HCT541 Y7 from the ATF22V10 arbitration output (Section 1.2): Z80 RD# during CPU ownership, Pico DMA read signal during DMA ownership.
	CE#	Pin 22	Input. Active Low. Driven by SN74HCT541 Y8 from the ATF22V10 arbitration output (Section 1.2): Z80 MREQ# during CPU ownership so I/O cycles cannot access SRAM, Pico DMA chip-enable signal during DMA ownership.
	CE2	Pin 30	Input. Active High. Tie permanently to VCC (+5V) to enable the device through the active-low CE# input.

1.1 Z84C0020PEC-to-AS6C1008-55PCN Wiring

The following is the direct run-mode connection verified against the manufacturers' 40-pin PDIP and 32-pin PDIP pin assignments. The AS6C1008 is a 128K x 8 device, so A16 must be tied LOW to expose its lower 64KB as the Z80's address space. CE2 must be tied HIGH, and pin 1 is not connected.

Manufacturer datasheets:

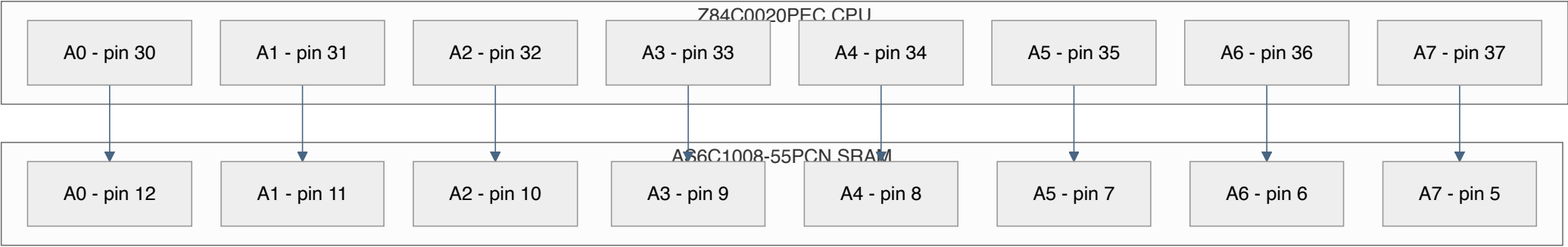
- [Zilog Z84C00 CMOS Z80 CPU Product Specification](#)
- [Alliance Memory AS6C1008 128K x 8 Low Power CMOS SRAM Datasheet](#)



SOP/P-DIP

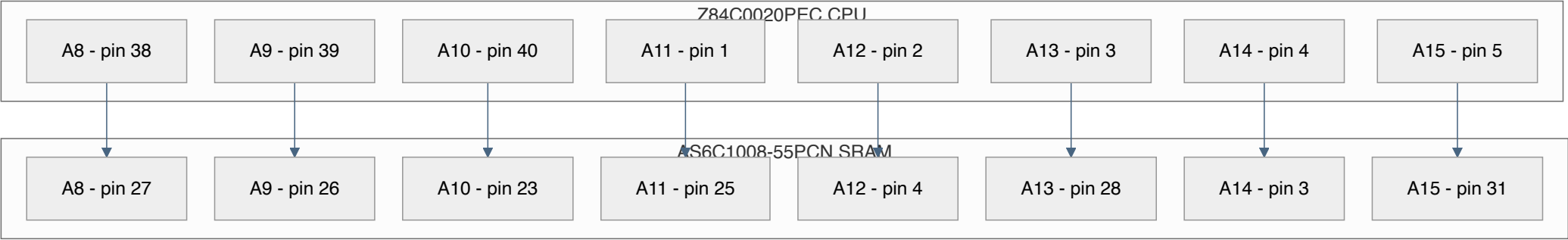
Address Bus A0-A7

Address Bus A0-A7



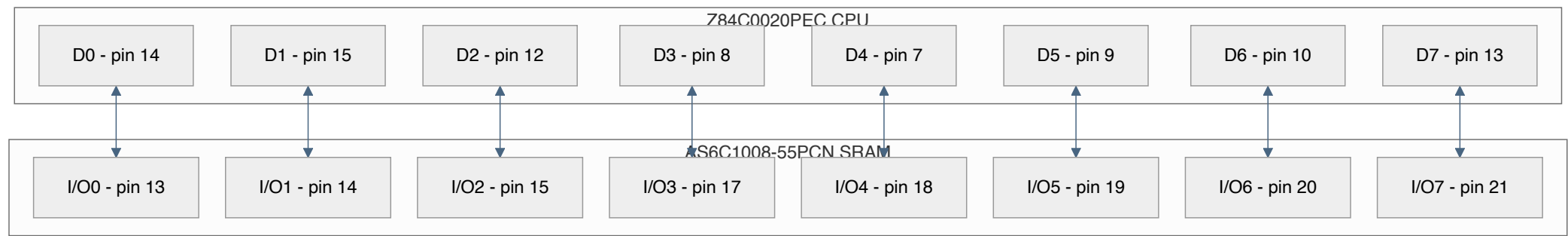
Address Bus A8-A15

Address Bus A8-A15



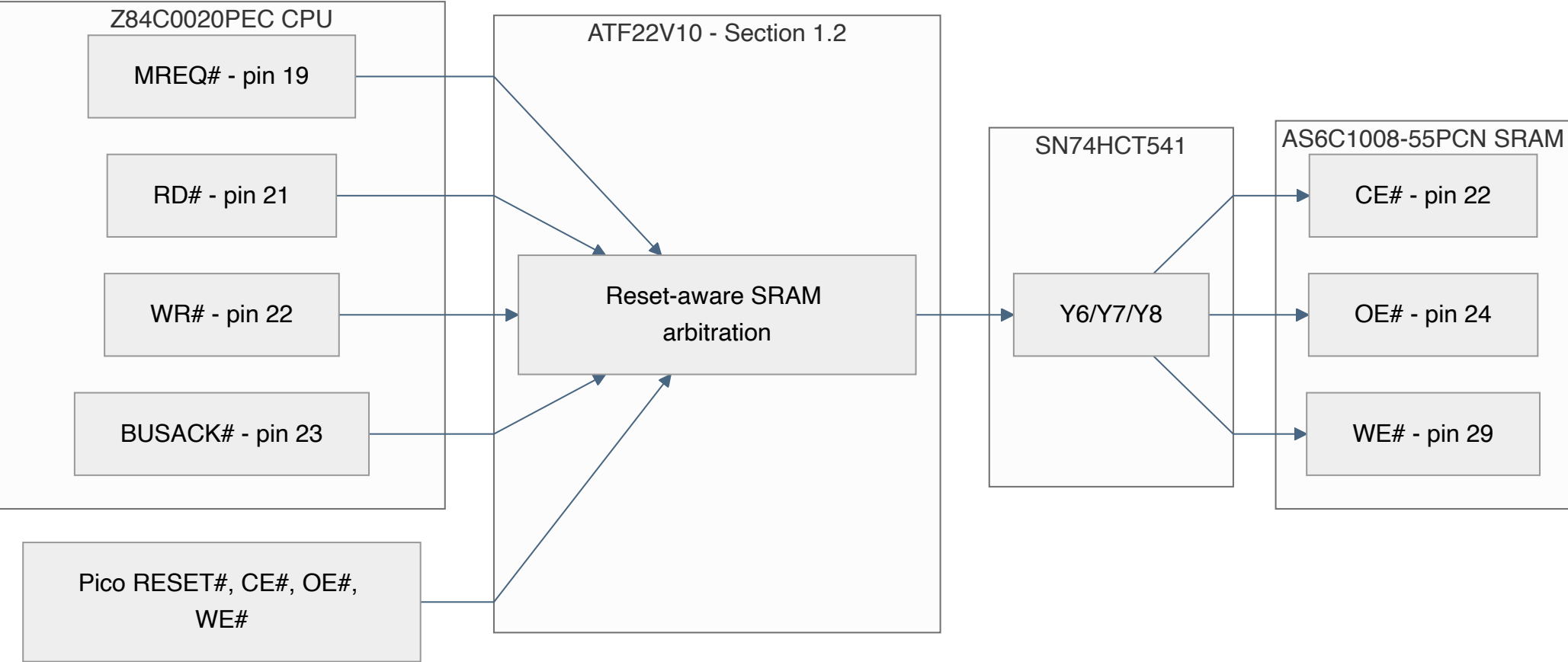
Data Bus D0-D7

Data Bus D0-D7



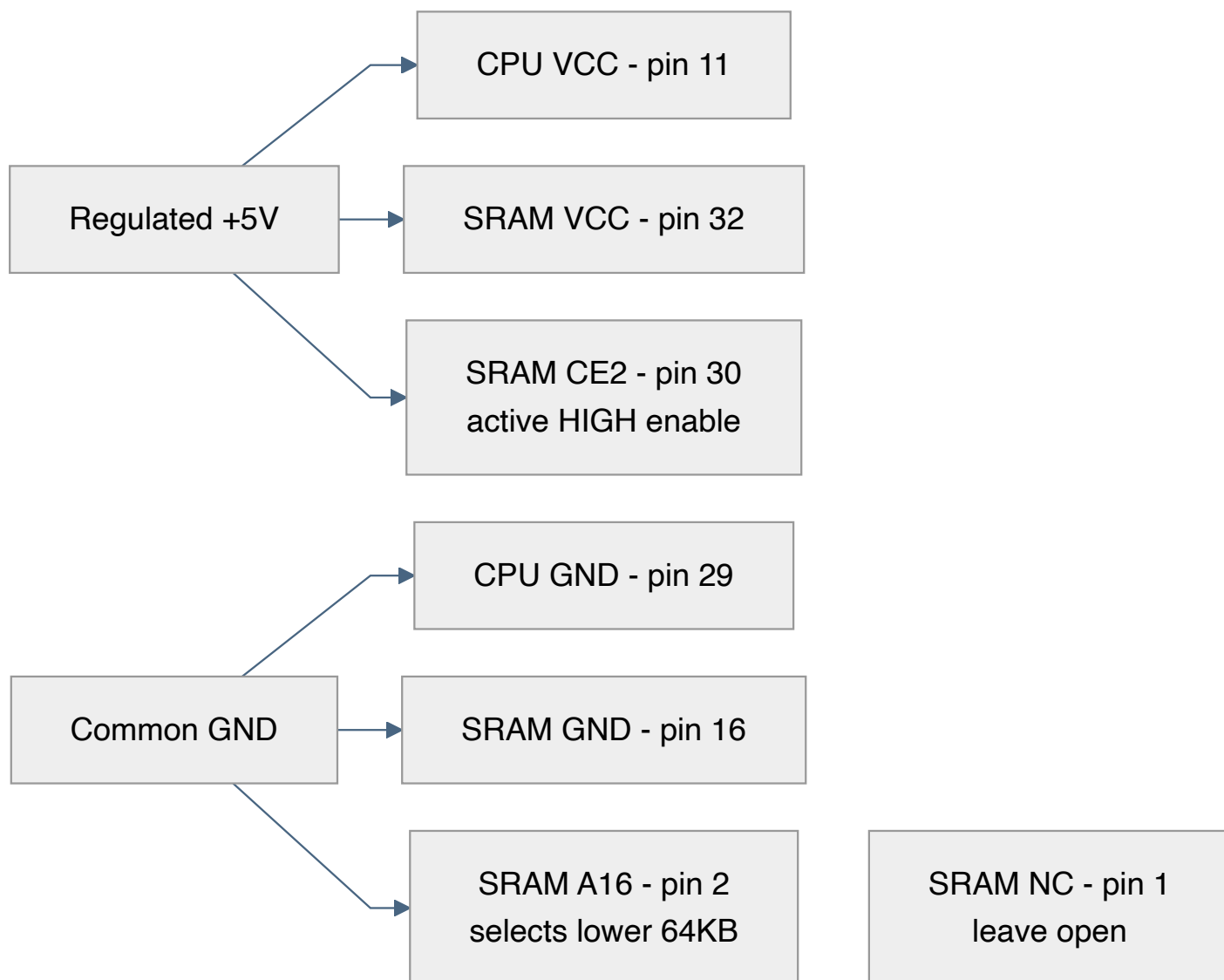
Memory Control

Memory Control



Power and Fixed Pins

Power and Fixed Pins



SRAM control-source arbitration: MREQ#/RD#/WR# above are the CPU-owned run-mode inputs of the ATF22V10 described in Section 1.2, never directly to the SRAM. The programmed equations select either these inputs or the Pico DMA controls; the three results then pass through dedicated SN74HCT541 channels to the SRAM.

1.2 SRAM Control-Source Arbitration: ATF22V10B/C

The Z80 and Pico must never drive SRAM CE#, OE#, or WE# simultaneously. The ATF22V10 implements three combinational 2:1 selectors with a shared ownership condition:

```
CPU_OWNS_SRAM = RESET# AND BUSACK#

SRAM_WE_PRE# = CPU_OWNS_SRAM ? Z80_WR# : PICO_WE#
SRAM_OE_PRE# = CPU_OWNS_SRAM ? Z80_RD# : PICO_OE#
SRAM_CE_PRE# = CPU_OWNS_SRAM ? Z80_MREQ# : PICO_CE#
```

RESET#	BUSACK#	Selected source	Required state
0	X	Pico	Held-reset DMA or Z80 absent
1	0	Pico	BUSREQ#/BUSACK# DMA ownership
1	1	Z80	Normal CPU execution

This covers the two cases that BUSACK# alone cannot distinguish. During RESET the Z80 drives BUSACK# inactive HIGH, and with the Z80 physically absent its pull-up also reads HIGH. RESET# LOW therefore selects the Pico in both cases. Only an out-of-reset CPU that has not granted the bus can select MREQ#/RD#/WR#.

The PLD source adds the logically redundant consensus term Z80_CONTROL# & PICO_CONTROL# to each output. It does not change this truth table, but it holds an inactive HIGH continuously while RESET# or BUSACK# transfers ownership with both candidate controls HIGH. Without that term, unequal GAL product-term delays can create a static-1 hazard, seen externally as a short active-low SRAM control pulse. Each hardened equation uses four product terms, within every ATF22V10 macrocell's capacity.

Program the selected ATF22V10B or ATF22V10C with [src/pld/sram_control.pld](#). Use the exact device algorithm listed by the programmer, write the generated JEDEC file, read it back, and require a verify pass before installation. Do not enable the optional power-down mode or security fuse.

ATF22V10 pin	Signal	Connection
1	RESET#	Pico GP3 and Z80 pin 26 direct node
2	BUSACK#	Z80 pin 23
3	PICO_WE#	Pico GP22
4	Z80_WR#	Z80 pin 22
5	PICO_OE#	Pico GP26
6	Z80_RD#	Z80 pin 21
7	PICO_CE#	Pico GP5
8	Z80_MREQ#	Z80 pin 19
9	DATA_ENABLE	Pico GP7; 10 kOhm pull-down to GND
10	ADDR_ENABLE	Pico GP9; 10 kOhm pull-down to GND
13	Unused input	Tie to GND
11	DATA_DIR	Pico GP6; 10 kOhm pull-down to GND
12	GND	Common ground

ATF22V10 pin	Signal	Connection
14	SRAM_WE_PRE#	SN74HCT541 A6 pin 7
15	SRAM_OE_PRE#	SN74HCT541 A7 pin 8
16	SRAM_CE_PRE#	SN74HCT541 A8 pin 9
17	DATA_UP_OE#	SN74AHCT245N OE# pin 19
18	DATA_DOWN_OE#	SN74LVC245AN OE# pin 19
19	MCP_RESET_DRIVE	4.7 kOhm to Q1 base; Q1 collector drives MCP RESET#
20-23	Unused outputs	Program constant LOW; leave open
24	VCC	Regulated +5 V

The ATF22V10's guaranteed HIGH output is 2.4 V, which satisfies the HCT541's 2.0 V input threshold but not the AS6C1008's 5 V CMOS input threshold. Never bypass the HCT541 on these three paths. Fit 10 kOhm pull-ups at HCT541 A6-A8 and at the final SRAM CE#/OE#/WE# nodes. The input pull-ups keep the permanently enabled HCT541 deterministic if the GAL is missing; the output pull-ups keep the SRAM inactive if the HCT541 is physically missing. All installed source devices must be powered as required by Section 0.4.

The same 2.4 V guaranteed HIGH exceeds the 2.0 V input-HIGH minimum of both data-transceiver OE# inputs. Two additional equations use two product terms each:

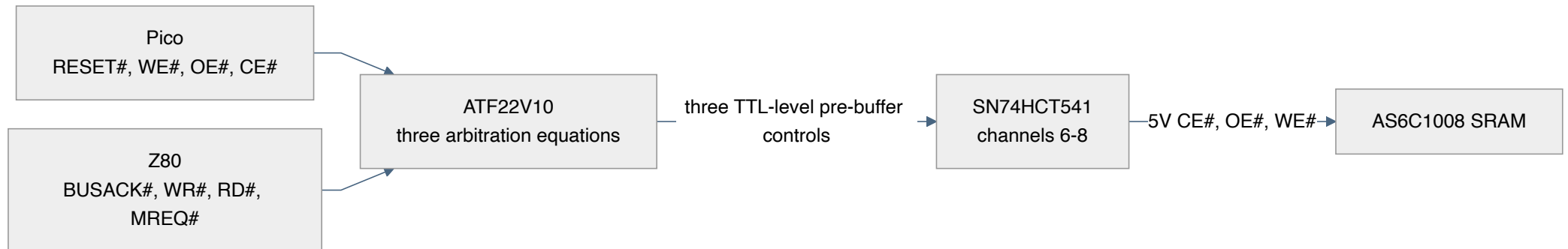
```
DATA_UP_OE#   = NOT DATA_ENABLE OR NOT DATA_DIR
DATA_DOWN_OE# = NOT DATA_ENABLE OR DATA_DIR
MCP_RESET_DRIVE = NOT ADDR_ENABLE
```

Both paths are disabled whenever DATA_ENABLE is LOW. When it is HIGH, exactly one path is enabled. Firmware changes DATA_DIR only while DATA_ENABLE is LOW, so the shared NOT DATA_ENABLE product term holds both OE# outputs inactive throughout the transition. SN74LVC245A Ioff permits its OE# pin to remain driven from the 5 V GAL while the LVC device's 3.3 V supply is absent.

For address isolation, connect GAL pin 19 through 4.7 kOhm to the base of Q1 (2N3904), add 47 kOhm base-to-emitter, ground the emitter, pull MCP RESET# up to 5 V through 10 kOhm, and connect the collector to RESET#. ADDR_ENABLE LOW makes GAL pin 19 HIGH and Q1 asserts reset, which forces all MCP GPIO registers back to input mode. ADDR_ENABLE HIGH releases reset; firmware then configures IODIR before using the address bus.

During Phase 6 with the Z80 absent, hold GP3 RESET# LOW. The programmed logic then selects the Pico controls regardless of the pulled-up, undriven BUSACK# input.

1.2 SRAM Control-Source Arbitration: ATF22V10B/C



2. 16-Bit Address Expansion Interface: MCP23S17-E/SP

The MCP23S17 acts as the dedicated 16-bit register shifter interfacing the Pico 2's SPI bus with the shared 5V address bus. During DMA block injection, it drives the target SRAM locations. During an active I/O trap, its GPIO ports remain inputs, allowing the MCP23S17 to monitor the address bus states driven by the frozen Z80 CPU.

MCP23S17 Pin Designation	Target Connection	System Logic Role
GPA0 – GPA7 (Port A)	Shared A0-A7 directly	Lower address byte drive/sample
GPB0 – GPB7 (Port B)	Shared A8-A15 directly	Upper address byte drive/sample
CS# (Pin 11)	SN74HCT541 Y3 pin 16, from Pico GP21 via A3	SPI Hardware Chip Select (Active Low), 5 V translated
CLK / SCK (Pin 12)	SN74HCT541 Y4 pin 15, from Pico GP18 via A4	SPI Master Clock Train Input, 5 V translated
SI (Pin 13)	SN74HCT541 Y5 pin 14, from Pico GP19 via A5	SPI Master-Out-Slave-In (MOSI Path), 5 V translated
SO (Pin 14)	SN74LVC244AN channel 2A1/2Y1 to Pico 2 GP20	SPI Master-In-Slave-Out (MISO Path), buffered from 5 V to 3.3 V; fit the Section 0.3 pull-up because SO is high-impedance while CS# is HIGH
A0, A1, A2 (Pins 15-17)	Tied to GND	Hardware address = 000; matches the fixed 0x40/0x41 opcode used in firmware regardless of the IOCON.HAEN state, and prevents floating address-select inputs
RESET# (Pin 18)	10 kOhm pull-up to 5 V and Q1 collector	LOW outside address access; reset default makes all GPIO inputs

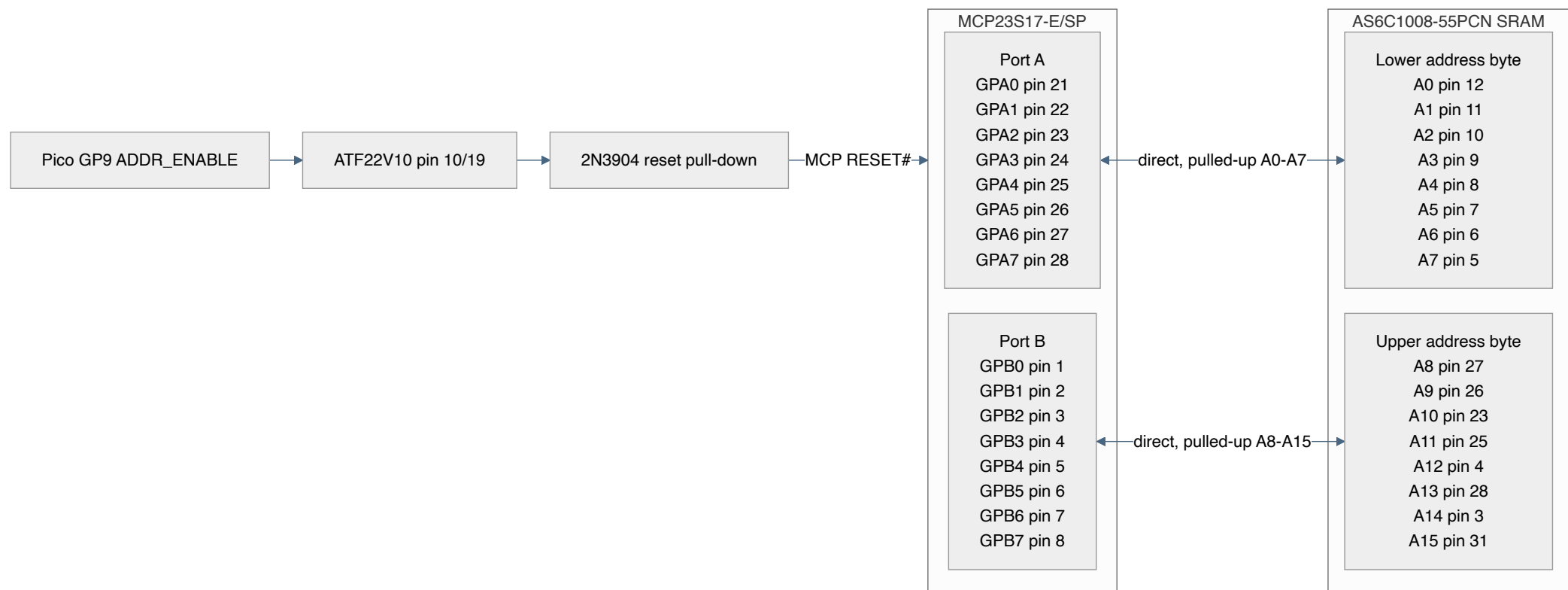
[Microchip MCP23017/MCP23S17 16-Bit I/O Expander with Serial Interface Datasheet](#)

2.1 MCP23S17-to-SRAM Address Wiring

The MCP23S17 connects directly to the shared address bus: GPA0-GPA7 to A0-A7 and GPB0-GPB7 to A8-A15. Fit one 8x10 kOhm bussed pull-up network per byte, with each common pin at 5 V. The Z84C00 guarantees 4.2 V at its light-load VOH2 point against the MCP's 4.0 V input minimum; the external pull-ups improve the static HIGH level and add at most about 0.5 mA load per LOW bit. MCP GPIO outputs guarantee at least 4.3 V at 5 V, exceeding the SRAM's 3.5 V input minimum.

Hardware isolation comes from RESET#, not a bus transceiver. GP9 ADDR_ENABLE feeds GAL pin 10. When LOW, GAL pin 19 drives Q1 and holds MCP RESET# LOW, forcing IODIRA/IODIRB to their all-input reset values. When HIGH, Q1 releases RESET#; firmware waits, then programs OLAT before changing IODIR to outputs. It reasserts reset before releasing the Z80. GP9 has a 10 kOhm pull-down, so Pico reset or power loss fails closed.

2.1 MCP23S17-to-SRAM Address Wiring



3. Physical Partitioning & Breadboard Topology

The layout enforces a strict three-zone model across three 830-point breadboards to minimize cross-talk and propagation delay across the distinct 3.3V and 5V power domains. Each zone below lists the specific chips to place on that board and where to seat them.

Place the three BB830s side by side with their long edges parallel: Memory on the left, Core in the center, and Peripheral on the right. Rows 1-63 run in the same direction on all three boards, so equal row numbers align across both board boundaries. The Core Board's central position is deliberate: the dominant Memory/Core paths are the shared 16-bit address and 8-bit data buses plus the MCP23S17 address/SPI interface, while the dominant Core/Peripheral paths are the Pico's 8-bit data interface and supervisor controls. Seven Pico-to-GAL signals span both board boundaries without a buffer: PICO_CE#, PICO_OE#, PICO_WE#, RESET#, DATA_ENABLE, DATA_DIR, and ADDR_ENABLE. RESET# is a three-way node also tapped by the Z80 on the Core Board; the others simply cross the Core Board without connecting to a Core component. MCP SO also crosses both boundaries to the Peripheral Board's LVC244. Keep these low-activity paths away from CLK. Putting either outer cluster in the center would shorten these four wires only by forcing one of the much wider address/data interfaces to span two board widths.

- **Memory Board (Left Zone):** AS6C1008-55PCN SRAM, the programmed ATF22V10, and MCP23S17. The GAL's four CPU-side inputs (BUSACK#, MREQ#, RD#, WR#) cross from the Core Board, and its three pre-buffer outputs cross back to the Core Board's SN74HCT541 before the buffered result returns here as SRAM CE#/OE#/WE#. This deliberate round trip keeps the HCT541's clock channel local to the Z80 instead (see below), which matters far more than SRAM control length since SRAM CE#/OE#/WE# only toggle at the Z80 bus-cycle rate, the same class as MREQ#/RD#/WR#. The MCP23S17 now sits here so its 16 port lines join the pulled-up shared address bus at the Memory/Core boundary. Q1 beside it provides reset-based hardware isolation. HCT541 SPI outputs also cross only the Memory/Core boundary. Its SO output is the one MCP signal that continues across Core to the LVC244. Row budget: 16 (SRAM) + 12 (ATF22V10) + 14 (MCP23S17) = 42 of 63 rows.
- **Core Board (Center Zone):** Z84C0020PEC CPU and the SN74HCT541N output buffer. Keep the HCT541 beside the Z80 so its Y1 clock output never crosses a board boundary, the same placement rule the discrete design used for its dedicated clock buffer. The Z80 address pins join the SRAM/MCP address trunk directly. The photographed plug-in supply reserves rows 1-3, leaving 60 usable terminal rows. The chip budget is 20 (Z80) + 10 (SN74HCT541N) = 30 of those 60 rows, leaving 30 rows for socket-body clearance, decoupling, and wiring. *No hardware wait-state latches or flip-flops are used.*
- **Peripheral Board (Right Zone):** Raspberry Pi Pico 2, SN74LVC244AN monitor buffer, SN74AHCT245N upward data path, and SN74LVC245AN downward data path. The ATF22V10 on Memory provides the mutually exclusive enables. Rotate both data transceivers so their B-port pin rows face the Core Board and their A-port pin rows face the Pico; tie AHCT DIR HIGH and LVC DIR LOW. Orient the LVC244 so its 1A input pin row faces the Core Board and its 1Y output pin row faces inward toward the Pico; its fifth input, MCP SO, arrives from Memory across Core. Row budget: 20 (Pico) + 10 (LVC244) + 10 (AHCT245) + 10 (LVC245) = 50 of 63 rows.

The following row-aligned schedule was selected by evaluating every valid per-board package ordering and gap distribution against grouped signal-count weights, iterating the comparison across all three boards. This is a routing aid, not an instruction to equalize individual wire lengths:

Board	Terminal-row schedule	Unallocated rows
Memory	Unallocated 1-4; GAL 5-16; gap 17; SRAM 18-33; gap 34; MCP23S17 35-48	1-4 and 49-63 (19)
Core / middle	Supply clearance 1-3; unallocated 4-7; HCT541 8-17; gap 18; Z80 19-38	4-7 and 39-63 (29)
Peripheral	Pico 1-20; gap 21; LVC244 22-31; gap 32; AHCT245 33-42; gap 43; LVC245 44-53	54-63 (10)

3.1 Package Orientation and Pin 1

Use this convention for both the table and the placement image: view each BB830 from above and rotate it 90 degrees counter-clockwise from the manufacturer's landscape drawing, so **row 1 is at the top, row 63 is at the bottom, A-E are on the left, and F-J are on the right**. Memory, Core, and Peripheral then sit left-to-right. Do not mirror any board.

Seat every DIP socket across the center ravine before wiring it. A 0.3-inch DIP uses the E/F holes immediately beside the ravine. The Z80 and SRAM are 0.6-inch-wide DIPs: dry-fit their specified 0.6-inch sockets in two terminal columns matching the actual 15.24 mm lead-row span; do not force or bend them into E/F. Both pin rows must remain on opposite, electrically isolated sides of the ravine. Install the IC only after marking the socket's pin-1 corner and matching the IC notch or dot to the socket.

Board / device	Occupied rows	Body orientation	Pin 1 location (top view)	Opposite corner check
Memory / ATF22V10	5-16	Notch toward row 1	E5	Pin 24 at F5
Memory / AS6C1008 SRAM	18-33	Notch toward row 1	A-E pin-row side at row 18	Pin 32 on F-J side at row 18
Memory / MCP23S17	35-48	Notch toward row 63	F48	Pin 28 at E48
Core / SN74HCT541	8-17	Notch toward row 63	F17	Pin 20 at E17; Y1 pin 18 is then close to Z80 CLK
Core / Z84C0020	19-38	Notch toward row 1	A-E pin-row side at row 19	Pin 40 on F-J side at row 19
Peripheral / Pico 2 W	1-20	USB connector toward row 1	Header pin 1 on A-E side at row 1	Header pin 40 on F-J side at row 1
Peripheral / SN74LVC244	22-31	Notch toward row 1	E22	Pin 20 at F22; 1A inputs face Core
Peripheral / SN74AHCT245	33-42	Notch toward row 63	F42	Pin 20 at E42; B1-B8 face Core
Peripheral / SN74LVC245	44-53	Notch toward row 63	F53	Pin 20 at E53; B1-B8 face Core

For the non-DIP keyed parts:

- **Q1 (2N3904):** place beside the MCP, not across the ravine. TO-92 lead order can vary by manufacturer; use the purchased part's datasheet to identify emitter/base/collector and mark E–B–C beside its holes. Do not rely on flat-face orientation alone.
- **RN1/RN2:** place as single-row SIPs near SRAM/MCP, parallel to the ravine, with the dot/common pin toward row 1 and wired to +5 V.
- **RN3:** place beside the AHCT/LVC A-port node with its dot/common pin toward row 1 and wired to GND.
- **1N5819 and electrolytics:** the diode band faces Pico VSYS; every electrolytic + lead goes to its positive rail. Mark polarity on the breadboard before insertion.
- **Plug-in supply module:** orient it only from its printed +, –, input, and output labels, then confirm every rail with a meter while unloaded. There is no generic module orientation; never infer polarity from USB-jack position or board color.

Before applying power, inspect every socket from above and verify the pin-1 location against this table. Then use continuity mode to prove that opposite-side pins at the same row are not shorted through a terminal strip.

This placement uses row alignment to reduce diagonal jumper length:

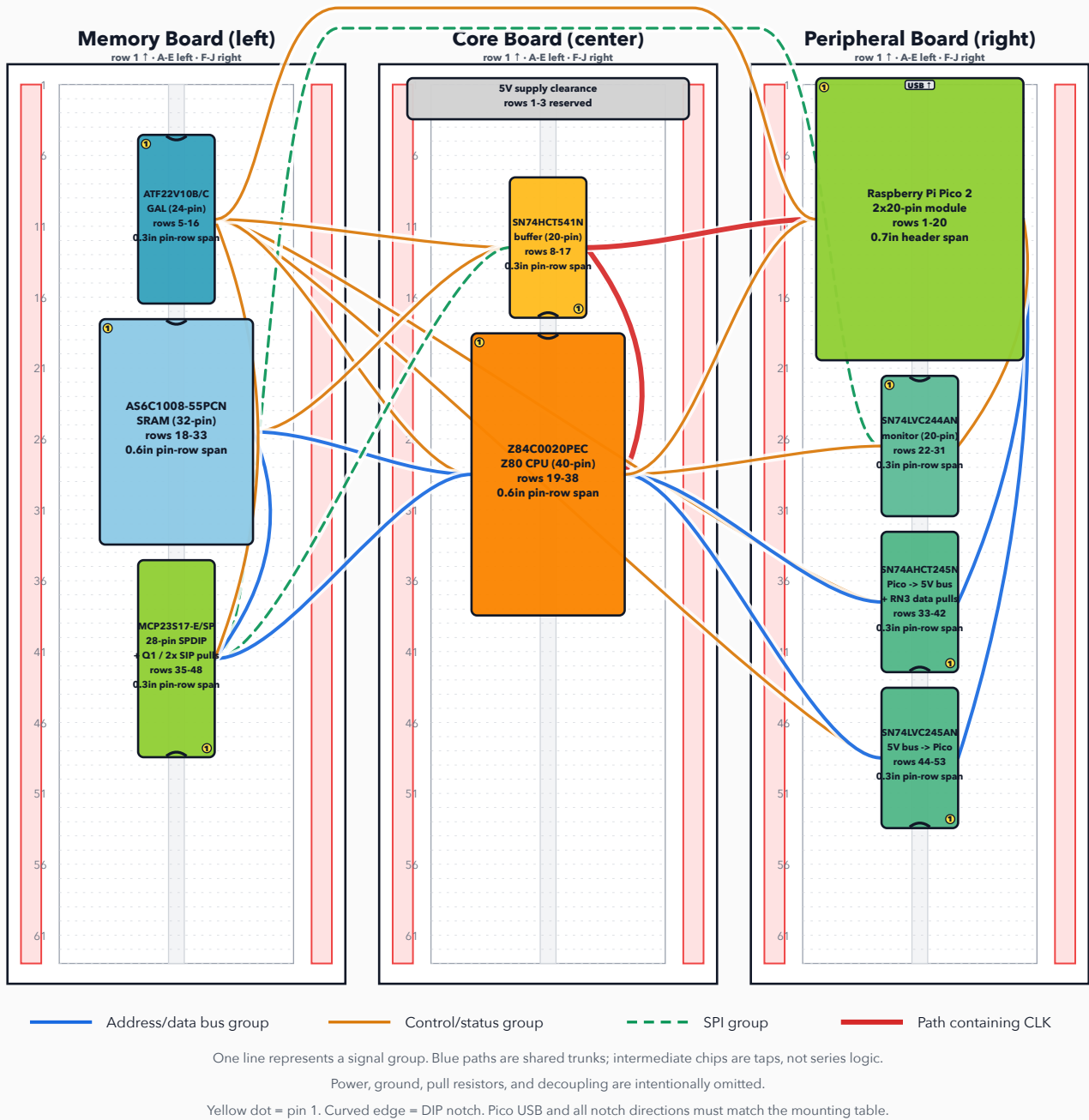
- **Memory/Core:** SRAM rows 18-33 overlap the Z80's rows 19-38 for the full A0-A15/D0-D7 trunk. GAL rows 5-16 overlap the HCT541, while MCP23S17 rows 35-48 meet the lower end of the direct Z80 address trunk.
- **Core/Peripheral:** Pico rows 1-20 overlap HCT541 and the top of the Z80. LVC244 rows 22-31 overlap the Z80 monitor sources. Both data transceivers overlap the lower Z80 region while remaining directly adjacent to the Pico on the same board.
- **Peripheral/Memory:** Pico rows 1-20 overlap GAL rows 5-16, reducing the vertical component of the seven Pico-to-GAL paths and two OE# returns. MCP SO remains one grouped low-activity crossing to the LVC244.

This schedule includes one empty row between every socket or module and still fits all three boards. Mark the actual supply-module overhang and socket outlines on the empty boards before wiring; the schedule is not valid if the supply consumes more than the assumed three rows.

BB830 physical placement and grouped chip connections

Boards are side by side with long edges parallel; equal terminal-row numbers align laterally

GAL controls data OE# and MCP RESET#; MCP ports connect directly to pulled-up A0-A15



Use one supply-entry point and fan out +5 V and GND to each board; do not daisy-chain the boards' power rails end-to-end. Run the Pico 3.3 V rail separately to the two LVC devices. Bond adjacent boards with multiple short ground jumpers, especially beside the address/data bus crossings and CLK. Verify every BB830 distribution rail end-to-end with a meter before fitting links; never assume visually aligned rail segments are internally continuous.

3.2 KiCad Electrical Schematic

The native KiCad 10 schematic is the electrical source of truth for pin numbers, named nets, explicit no-connect markers, and ERC. It uses project-local symbols so the exact ATF22V10, Z80, SRAM, Pico, and translator pin definitions travel with the repository. RESET# is drawn as an explicit three-way Pico/Z80/GAL wire junction. Address, data, SPI, monitor, and control paths are shown as unfolded KiCad vector or group buses with every member named; matching labels on the chip-pin stubs provide the exact electrical connections without implying that signals pass through an intermediate chip as series logic.

Artifact	Purpose
KiCad project and native schematic	Editable KiCad 10 source
Project symbol library	Exact local pin names, numbers, and ERC electrical types
SVG export and PDF export	Zoomable full schematic renderings
KiCad netlist and independent net manifest	Machine-readable connectivity
ERC report	KiCad 10.0.5 result: zero violations with errors, warnings, and exclusions included

The schematic contains 59 physical components, including 31 discrete resistors, three SIP networks, and 12 fitted capacitors, plus one nonphysical #FLG01 power marker used only by ERC. KiCad's exported netlist matches the independently generated manifest exactly: 79 real nets and 348 component pin endpoints. The ERC-only power marker and KiCad's synthetic no-connect nets are excluded from that comparison.

To regenerate and validate the native source, exports, strict ERC report, and independent netlist comparison:

```
python3 -m venv .venv-kicad
.venv-kicad/bin/pip install -r scripts/requirements-kicad.txt
PYTHON=.venv-kicad/bin/python npm run kicad
```

The command runs [build-kicad-schematic.py](#), KiCad CLI upgrade/export/ERC, and [check-kicad-netlist.py](#). Any ERC violation or net/endpoint mismatch fails the build.

3.3 High-Speed Interconnect Routing

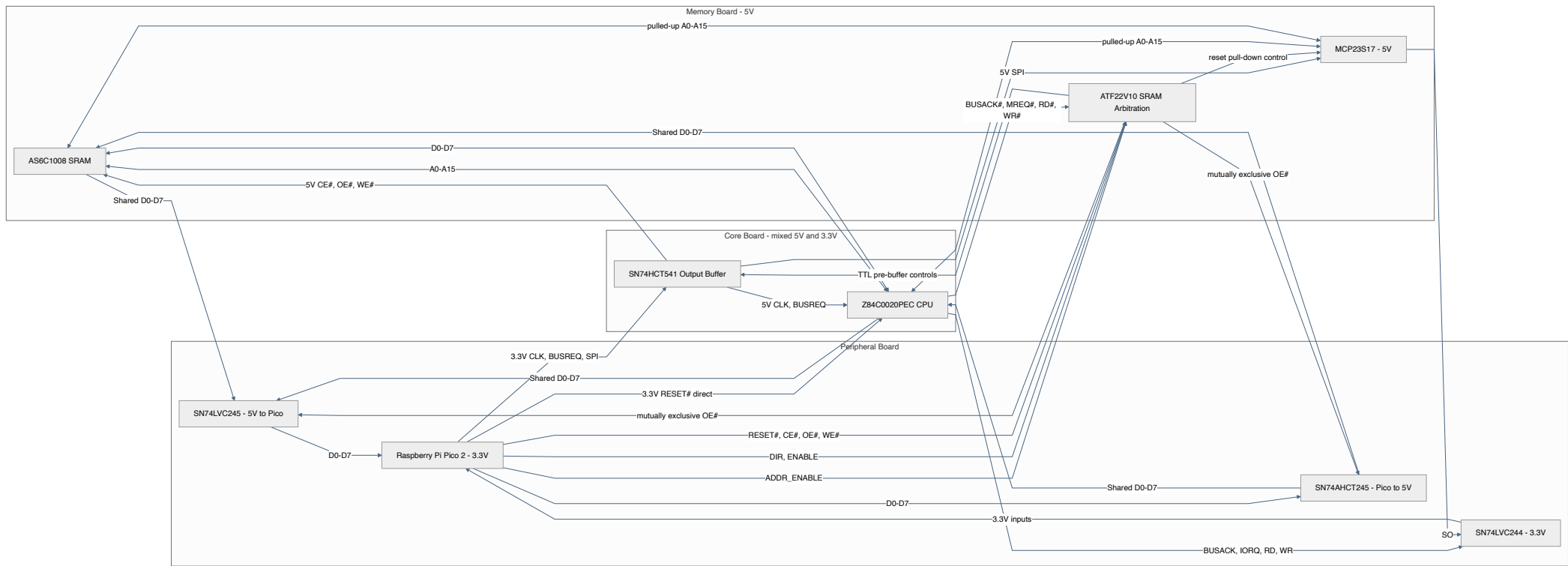
At the qualified 1-4 MHz clock rates, propagation skew from a few millimetres of wire-length difference is negligible compared with the Z80 timing budget. Solderless-breadboard reliability is instead dominated by total wire length, stubs, loop area, contact resistance, and fast-edge ringing. Route each bus as a short grouped trunk with roughly similar paths, but do not add serpentine wire merely to make lengths equal:

- **A0-A15 (hard construction rule):** keep the Z80, SRAM, MCP23S17, and both SIP pull-up networks on one short common trunk. Do not build this as a star and do not leave long branches to any device. A layout that cannot satisfy this rule fails the placement review and must be rearranged before wiring the remaining signals.
- **D0-D7:** join both Peripheral data-transceiver B ports to one short 5 V trunk crossing to the Core Board, then continue that shared trunk through the Z80 to the adjacent SRAM. Keep the two A-port taps to the Pico short and parallel; never route one translator through the other.
- **SN74HCT541 Y6-Y8 to SRAM CE#/OE#/WE#:** these cross from the Core Board's HCT541 to the Memory Board's SRAM; route them as one grouped trunk at the board boundary. Exact length matching is unnecessary.
- **CLK (SN74HCT541 Y1 pin 18 to Z80 pin 6):** route this as the single shortest, most direct jumper, kept away from the address bus. Do not lengthen it to match other nets; clock is the most edge-rate-sensitive signal in the design. This is why the HCT541 is seated on the Core Board rather than beside the GAL.

Route a ground jumper alongside every inter-board signal group and add ground probe points near CLK, IORQ#, MREQ#, RD#, WR#, SRAM CE#/OE#/WE#, and each bus transceiver. Keep all jumpers as short as the placement allows.

3.4 Major Chip Interconnection Overview

3.4 Major Chip Interconnection Overview



4. Output Buffer Mapping: SN74HCT541N (DIP-20)

The 5 V-powered SN74HCT541 provides all eight required high-level outputs. Its TTL-compatible inputs accept both Pico 3.3 V signals and the ATF22V10's guaranteed 2.4 V HIGH. At the light CMOS loads used here, its 5 V outputs satisfy the Z80 clock's strict V_{IHC} threshold, the MCP23S17's $0.8V_{DD}$ SPI threshold, and the SRAM's CMOS control-input threshold. Its worst-case propagation delay is 29 ns at 4.5 V with a 50 pF load. Combined with the ATF22V10C-15's 15 ns combinational delay and the SRAM's 55 ns chip-enable access, the control-to-data path can approach 99 ns before breadboard interconnect allowance. This is why 1-4 MHz is measured rather than assumed, and why this design is not a 20 MHz system despite using a 20 MHz-rated CPU.

Channel	HCT541 input	HCT541 output	Function
1	A1 pin 2 from Pico GP2	Y1 pin 18 to Z80 CLK pin 6	Master clock
2	A2 pin 3 from Pico GP4	Y2 pin 17 to Z80 BUSREQ# pin 25	DMA request
3	A3 pin 4 from Pico GP21	Y3 pin 16 to MCP23S17 CS# pin 11	SPI chip select
4	A4 pin 5 from Pico GP18	Y4 pin 15 to MCP23S17 SCK pin 12	SPI clock
5	A5 pin 6 from Pico GP19	Y5 pin 14 to MCP23S17 SI pin 13	SPI MOSI
6	A6 pin 7 from ATF22V10 pin 14	Y6 pin 13 to SRAM WE# pin 29	Arbitrated write enable
7	A7 pin 8 from ATF22V10 pin 15	Y7 pin 12 to SRAM OE# pin 24	Arbitrated output enable
8	A8 pin 9 from ATF22V10 pin 16	Y8 pin 11 to SRAM CE# pin 22	Arbitrated chip enable

Tie both active-low output enables, OE1# pin 1 and OE2# pin 19, to GND. Connect VCC pin 20 to regulated +5 V and GND pin 10 to common ground. The buffer is permanently enabled; the Pico-side defaults and GAL equations therefore establish safe inactive outputs before firmware enables its GPIO drivers.

Pico GP3 RESET# bypasses the HCT541 and connects directly to Z80 pin 26 and ATF22V10 pin 1. Its 3.3 V HIGH exceeds the Z84C00's 2.2 V and the ATF22V10's 2.0 V input-HIGH minima. Keep its 10 kOhm pull-down to GND and never fit a 5 V pull-up on this node.

5. Transceiver Operating Modes & Isolation Tables

The transceivers isolate the supervisor elements (Pico 2 and MCP23S17) from the main bus during standard execution, preventing bus contention.

5.1 All-PDIP Data-Bus Translation and Interlock

No single PDIP part provides an SN74LVC8T245-equivalent combination of dual supplies, bidirectional translation, deterministic direction, three-state isolation, and partial-power-down safety. The breadboard design therefore uses two fixed-direction transceivers:

- **SN74AHCT245N (5 V):** A port is Pico D0-D7, B port is the 5 V bus, and DIR pin 1 is tied HIGH. Its TTL-compatible A inputs accept 3.3 V and its B outputs provide full 5 V CMOS levels.
- **SN74LVC245AN (3.3 V):** A port is Pico D0-D7, B port is the 5 V bus, and DIR pin 1 is tied LOW. Its B inputs tolerate 5.5 V, A outputs stay in the Pico domain, and Ioff prevents back-powering while 3.3 V is absent.

Bus bit	Pico GPIO / both A ports	5 V bus / both B ports
D0	GP10 / A1 pin 2	B1 pin 18
D1	GP11 / A2 pin 3	B2 pin 17
D2	GP12 / A3 pin 4	B3 pin 16
D3	GP13 / A4 pin 5	B4 pin 15
D4	GP14 / A5 pin 6	B5 pin 14
D5	GP15 / A6 pin 7	B6 pin 13
D6	GP16 / A7 pin 8	B7 pin 12
D7	GP17 / A8 pin 9	B8 pin 11

Connect both pin 10s to GND. Connect AHCT pin 20 to 5 V and LVC pin 20 to 3.3 V. Each device needs its own local 100 nF capacitor.

The existing ATF22V10 provides the hardware interlock using spare pins and product terms. Connect GP7 DATA_ENABLE to GAL pin 9 and GP6 DATA_DIR to pin 11. GAL pin 17 drives AHCT245 OE# pin 19; pin 18 drives LVC245 OE# pin 19. The two equations are documented in Section 1.2. No 5 V output drives GP6/GP7: they are GAL inputs with 10 kOhm pull-downs. During Pico power-off both inputs read LOW, so both GAL OE# outputs are HIGH. The LVC245's Ioff specification protects its 3.3 V-powered side while GAL pin 18 remains at a 5 V-domain HIGH.

System state	DATA_ENABLE GP7	DATA_DIR GP6	AHCT OE#	LVC OE#	Result
Isolated / run	0	X	1	1	Both Pico data paths high-impedance
DMA write	1	1	0	1	Pico → AHCT245 → 5 V bus
Readback / OUT trap	1	0	1	0	5 V bus → LVC245 → Pico

Firmware always drives DATA_ENABLE LOW, waits, changes DATA_DIR, waits, and only then drives DATA_ENABLE HIGH. The GAL truth table also makes simultaneous enables impossible for every static GP6/GP7 state. Do not substitute TXS0108E, TXB0108, BSS138, or resistor-divider breakouts: their automatic/pass-device behavior and loading assumptions are not equivalent to this controlled, multi-load push-pull bus.

5.2 Direct MCP23S17 Address-Bus Modes

System state	ADDR_ENABLE GP9	MCP RESET#	IODIRA/B	Functional role
DMA injection	1	1	0x00/0x00 after OLAT preload	MCP drives A0-A15

System state	ADDR_ENABLE GP9	MCP RESET#	IODIRA/B	Functional role
Trap address read	1	1	0xFF/0xFF	MCP samples the frozen Z80 address
Active execution / isolated	0	0	Reset default 0xFF/0xFF	MCP pins are inputs; Z80 owns A0-A15

Always assert ADDR_ENABLE LOW before releasing the CPU. Releasing MCP reset is not itself permission to drive: firmware must preload OLAT and hold Z80 RESET# or BUSACK# before writing IODIR outputs.

5.3 SN74LVC244AN 5 V-to-3.3 V Input Buffer

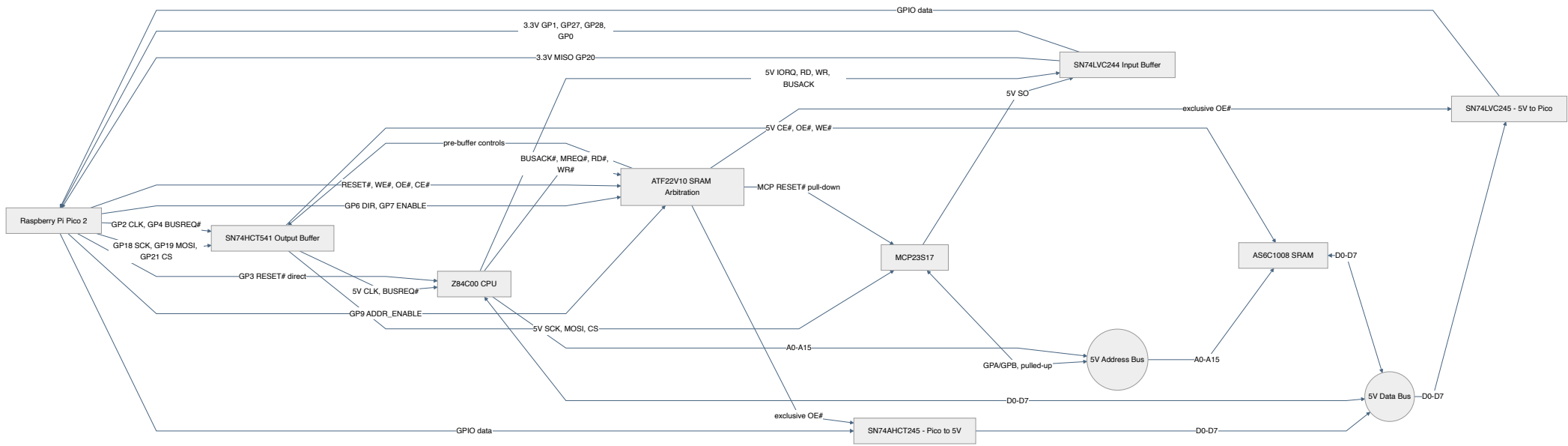
The RP2350's GP0-GP25 are 5 V-tolerant FT pads, but the 5.5 V rating applies only while IOVDD is powered at 3.3 V; with IOVDD at 0 V their absolute maximum is 3.63 V. GP26-GP29 are the QFN-60 package's ADC-capable pads and are not FT at all. Buffering all five incoming 5 V signals therefore avoids a power-sequencing constraint and keeps every Pico GPIO within its normal 3.3 V domain. Power the SN74LVC244AN from the Pico 3.3 V rail. Its inputs accept up to 5.5 V and its I_{off} specification protects both sides when VCC is 0 V. Tie both output enables (pins 1 and 19) to GND: every buffered signal -- BUSACK#, IORQ#, RD#, WR#, and MCP SO -- must always be readable, and none of them share a Pico GPIO with any other driver, so neither output enable needs gating. Wire the channels as follows; tie every unused input to GND and leave unused outputs open.

5 V source	LVC244 input	3.3 V output	Pico destination
Z80 BUSACK# pin 23	1A1 pin 2	1Y1 pin 18	GP0
Z80 IORQ# pin 20	1A2 pin 4	1Y2 pin 16	GP1
Z80 RD# pin 21	1A3 pin 6	1Y3 pin 14	GP27
Z80 WR# pin 22	1A4 pin 8	1Y4 pin 12	GP28
MCP23S17 SO pin 14	2A1 pin 11	2Y1 pin 9	GP20
GND	2A2/2A3/2A4 pins 13/15/17	2Y2/2Y3/2Y4 pins 7/5/3 open	Unused

VCC (pin 20) connects to 3.3 V and GND (pin 10) to common ground. The 5 V-side pull-ups in Section 0.3 keep every input defined when its source is absent or high-impedance.

5.4 Bus and Supervisor Signal Interconnections

5.4 Bus and Supervisor Signal Interconnections



6. Architectural Operational Boundaries

6.1 Synchronous Clock-Stop Trap Protocol

When the Z80 executes an I/O instruction, the falling edge of IORQ# trips a hardware edge interrupt on the Pico 2. The handler disables the hardware PWM clock slice after GPIO synchronization and interrupt-entry latency. Because the Z84C00 is fully static, the clock can then remain stopped indefinitely in either a HIGH or LOW state. The design has no hardware WAIT# or combinational clock gate, so safe trap frequency is a measured property, not an assumption.

6.2 Terminal I/O over Pico WebSocket

The final terminal is a virtual I/O peripheral implemented by the Pico, not a Z80-side UART. Z80 software performs ordinary IN and OUT instructions against a small terminal port pair; the Pico intercepts those cycles through the Section 6.1 clock-stop trap, then resumes the CPU after sampling or supplying the data byte. A browser connects to the Pico over Wi-Fi and receives the terminal stream through a WebSocket server running on the Pico 2 W.

The WebSocket server must run on the Pico's other core so Wi-Fi, lwIP, HTTP serving, and WebSocket polling cannot interfere with the timing of the Z80-facing supervisor path. Core 0 owns GPIO, MCP23S17 SPI, clock stop/resume, DMA, and the I/O trap. Core 1 owns Wi-Fi association, the embedded terminal page, WebSocket accept/send/receive, and network polling. The cores share terminal byte queues, one immutable disk-write queue, a small request/result pair for Z80 bus ownership, and atomic terminal/disk status words; they share no raw bus GPIO ownership.

Use the `pico-altair-8800` WebSocket console as the software pattern: one embedded HTML terminal page, one WebSocket server, and byte queues between the CPU-facing side and the network-facing side. The trap hook must never call lwIP, print, sleep, wait for a browser, or block on a queue. It may only enqueue Z80 output bytes, dequeue already-buffered browser input bytes, and report terminal status. Core 1 may drop, drain, or back-pressure network data; it must not take locks needed by the trap or touch any Z80 bus GPIO.

The initial terminal decode is intentionally small:

Port	Direction	Function
0x00	Z80 OUT	Terminal transmit byte from Z80 to browser
0x00	Z80 IN	Terminal receive byte from browser to Z80; returns 0x00 if empty
0x01	Z80 IN	Terminal status: bit 0 = receive byte available, bit 1 = transmit queue has room, bit 7 = WebSocket client connected

This is enough for ROM monitors, BASIC, CP/M console glue, and small diagnostics. If later software needs modem-control style signals, add more virtual status bits before adding hardware.

6.3 Onboard Flash CP/M Disk Storage

Boot images and CP/M disk images live in a reserved region of the Pico's own onboard flash, not on any external card or bus. This needs zero extra GPIO and zero extra parts: the same physical flash chip that already holds the Pico's own firmware is memory-mapped and directly readable by both cores at any time, so no SPI bus, socket, or level-shifted chip-select is involved. GP27 and GP28 keep their original job monitoring Z80 RD# and WR# (Section 5.3); nothing needs reclaiming for storage.

Capacity budget. The Pico 2 W's onboard flash is 4 MiB. Reserve the top 1.4375 MiB for a power-fail journal, a distinct Z80 boot package, and four complete 320 KiB disks, leaving 2.5625 MiB for Pico firmware. That still comfortably covers this project's code plus the Wi-Fi firmware/CLM blob embedded by `pico_cyw43_arch`.

Region	Flash offset (from flash base)	Size	Contents
Journal	0x290000	64 KiB	Eight rotating metadata/data sector pairs
Boot	0x2A0000	128 KiB	Manifest plus a maximum 64 KiB Z80 RAM image

Region	Flash offset (from flash base)	Size	Contents
Drive A	0x2C0000	320 KiB	CP/M system disk
Drive B	0x310000	320 KiB	CP/M data disk
Drive C	0x360000	320 KiB	CP/M data disk
Drive D	0x3B0000	320 KiB	CP/M data disk

Each 320 KiB slot is exactly eighty 4 KiB flash erase sectors with no partial sector left over, so every slot boundary is also a sector boundary.

Disk-image contract. Here, "320 KiB" means exactly 327,680 bytes: 2,560 linear 128-byte CP/M records. The default BIOS presents those as 80 logical tracks of 32 records, numbered 1-32, with no skew. This is a project-specific logical geometry, not the IBM 3740 8-inch SSSD format (77 tracks x 26 records x 128 bytes = 256,256 bytes); existing images in another 8-inch format must be converted rather than copied into a slot unchanged. The matching CP/M 2.2 DPB is SPT=32, BSH=4, BLM=15, EXM=1, DSM=155, DRM=63, AL0=0x80, AL1=0x00, CKS=0, and OFF=2. Keep these values in the Z80 BIOS and host image builder from one shared generated definition.

Reservation mechanism. Set the CMake variable PICO_FLASH_SIZE_BYTES to 0x290000 before `pico_sdk_init()`. In Pico SDK 2.2 this value sizes the generated linker's FLASH region, while the PICO_FLASH_SIZE_BYTES C macro still comes from `pico2_w.h` and must remain the physical 4 MiB. That deliberate same-name split lets the linker reject code or const data that grows into storage while `hardware_flash` still accepts physical offsets through 0x3FFFFFF. Do not pass `-DPICO_FLASH_SIZE_BYTES=0x290000` to the compiler: its flash erase/program bounds would then reject every storage write. Add `static_assert(PICO_FLASH_SIZE_BYTES == 4u * 1024u * 1024u, "Pico 2 W flash size changed")` to the storage module and inspect the link map in CI to require `__flash_binary_end <= 0x10290000`.

The build must also link `pico_flash`, `hardware_flash`, `hardware_watchdog`, `pico_multicore`, and `pico_util`. Define `PICO_FLASH_ASSUME_CORE1_SAFE=1`: core 0 uses `flash_safe_execute()` only for journal recovery before core 1 is launched, while later core-1 writes still lock out core 0 normally. Core 0 must never erase/program flash after core 1 starts. The essential CMake ordering is:

```
set(PICO_FLASH_SIZE_BYTES 0x290000)
pico_sdk_init()

target_compile_definitions(z80_supervisor PRIVATE
    PICO_FLASH_ASSUME_CORE1_SAFE=1)
target_link_libraries(z80_supervisor PRIVATE
    pico_flash hardware_flash hardware_watchdog pico_multicore pico_util)
```

Provisioning. Build each disk as a flat, exactly-320-KiB binary and build the boot package described below, then write them outside the running firmware. Use `-v` so `picotool` verifies every load:

```
picotool load -v -o 0x102A0000 -t bin z80boot.pkg
picotool load -v -o 0x102C0000 -t bin drive-a.img
picotool load -v -o 0x10310000 -t bin drive-b.img
picotool load -v -o 0x10360000 -t bin drive-c.img
picotool load -v -o 0x103B0000 -t bin drive-d.img
```

(0x10000000 is the Pico's XIP flash base address, so these absolute addresses are the flash-offset column above plus that base.) Reject any disk file whose host-side size is not exactly 327,680 bytes. If an RP2350 partition table is later embedded, declare these as data partitions and load them by partition ID; do not casually add `--ignore-partitions`. Firmware-only UF2 updates normally leave these addresses untouched, but `flash_nuke.uf2`, mass erase, or a replacement partition table destroys them, so keep host backups. There is no runtime bulk-image upload path in this design.

`z80boot.pkg` is little-endian. Its first 20 bytes are, in order, 32-bit magic 0x5442385A ("Z8BT"), 16-bit version 1, 16-bit header length 20, 32-bit image length, 32-bit IEEE CRC32 of the image, and 32-bit IEEE CRC32 of the preceding 16 header bytes. Pad the remainder of the first 4 KiB with 0xFF; the image begins at package offset 0x1000 and must be 1-65,536 bytes. The Z80 reset vector is at image offset zero. Generate this package and the BIOS DPB constants from one host tool so geometry and integrity metadata cannot drift.

Read path. A disk read is a plain pointer dereference into the memory-mapped flash region -- `XIP_BASE + drive_offset + sector_offset` -- with no SPI transaction, no DMA request, and no cross-core handshake. This is fast and safe to call directly from `io_trap_handler()` itself, unlike the deferred-queue pattern SD-based storage would have needed, provided no flash write (below) is concurrently in progress; the write path guarantees that by freezing the Z80 for its own duration, so a read trap can never race a write.

Write and recovery path. Updating one 128-byte CP/M record requires read-modify-erasing and reprogramming its enclosing 4 KiB flash block. The trap copies the complete 128-byte request into `disk_write_queue` and reports BUSY; it never erases flash itself. Core 1 constructs the new 4 KiB block in SRAM, asks core 0 to acquire BUSACK# while trapping remains enabled, and core 0 disables the trap only after BUSACK# is LOW. Core 1 then uses a bounded `flash_safe_execute()` call, which parks the registered core-0 victim and disables core-1 interrupts.

Each update rotates through one of eight 8 KiB journal pairs:

1. Erase the pair, then program the replacement 4 KiB data block.
2. Program a one-page header containing sequence, target offset, and CRC32. Until this valid header exists, the target remains untouched.
3. Erase/program the target block and verify all 4 KiB through XIP.
4. Erase the journal header only after verification succeeds.

At cold boot, core 0 scans valid journal headers before loading the Z80 image and restores them in sequence order. Therefore power loss before step 2 leaves the old target intact; power loss during or after step 3 leaves a valid replacement block from which boot recovery can finish. After a write, core 0 arms the trap while BUSACK# is still LOW and only then releases BUSREQ#, eliminating any interval in which the Z80 can run without I/O trapping. A bus-acquisition failure leaves the trap enabled; a release failure asserts RESET# and disables the trap. IORQ#, RD#, and WR# all tri-state whenever BUSACK# is asserted (Z8400 datasheet), so `io_trap_handler()` itself first confirms BUSACK# is still HIGH before touching anything; a stray edge while the Pico holds the bus disables the IORQ# interrupt and is ignored. The fitted 5 V-side 10 kOhm pull-ups in Section 0.3 keep the LVC244 inputs defined throughout the grant; Pico-side pulls cannot bias an input on the other side of the buffer. A successful write or recovery requires both a PIC0_0K safe-execute result and callback verification of the flash contents. A runtime safe-execute failure asserts RESET#, isolates the buses, stops CLK, and forces a watchdog reboot before attempting another core-0 request: SDK lockout-exit failure can leave core 0 parked, so continuing to its release queue would deadlock instead of recovering.

Flash write endurance is finite and chip-specific; confirm the Pico 2 W's actual onboard flash part's rated erase-cycle endurance before relying on this for a write-heavy workload. This partition suits occasional CP/M saves and file writes well; it is a poor fit for a disk used as constant scratch/swap space.

Core assignment. Reads happen inline in `io_trap_handler()` on core 0, since they need no cross-core coordination. Writes are deferred: the trap enqueues a request, and core 1 -- the same task that owns the WebSocket terminal (Section 6.2) -- checks that queue on every loop iteration before doing network work, asks core 0's foreground loop to freeze the Z80, performs the journaled update, then releases it and reports READY or ERROR. Wi-Fi association is a bounded polling state machine, so storage remains available with no access point. Flash access never touches SPI0, the MCP23S17, or any Z80 bus GPIO.

6.4 System Performance Envelope & Constraints

- **I/O Decode Width:** Strictly limited to **8-bit** decoding (monitoring address lines A0-A7 via the lower expander port).
- **Trap Latency Profile:** Only the interval from IORQ# falling to the final PWM edge determines whether the Z80 is frozen in time; the subsequent SPI servicing time is unrestricted while the static CPU is stopped. Measure that initial interval at every claimed clock rate. The design is suitable for low-rate virtual peripherals, not high-speed line tracing.
- **Clock Validation Targets:**
 - *Bring-Up Target:* **1 MHz**, qualified only after a logic-analyzer capture proves every I/O cycle stops before the Z80 samples or releases its data.
 - *Qualification Range:* **2 MHz – 4 MHz**, tested in the increments in Section 8.12. No rate in this range is guaranteed in advance.

- *Failure Boundary:* If the PWM cannot be stopped in time at a desired rate, add hardware WAIT#/clock gating rather than relying on faster firmware. Operation above 4 MHz is outside this design's qualification plan.
- *20 MHz CPU Rating:* The Z84C0020PEC rating applies to the CPU, not this no-wait-state breadboard system. At 20 MHz a clock period is 50 ns, shorter than the approximately 99 ns worst-case GAL + HCT541 + SRAM select-to-data path. Reaching 20 MHz requires a redesigned control path, hardware-generated memory and I/O wait states (or deterministic clock gating), and a PCB-level signal- integrity review; changing the Pico PWM frequency is insufficient.

7. Reference Firmware Implementations

The maintained, buildable firmware is the canonical implementation: [browse the source tree](#) or use the [complete source index](#). Section 8.13 provides design-level excerpts for the safety invariants and integration order: variable-frequency clock generation, bus acquisition, synchronous I/O trapping, flash image loading, and terminal integration. Do not copy those excerpts in place of the maintained source.

8. Progressive Build and Bring-Up Plan

Build and test one functional block at a time. Do not install the next chip until the current phase passes. Use sockets for all DIP devices, place a 100 nF ceramic capacitor directly across each IC's supply pins, and fit at least one 22 uF bulk capacitor per breadboard. Use a current-limited 5 V supply, multimeter, oscilloscope, and preferably a logic analyzer. Start each first power-up at a 100 mA current limit and remove power immediately if a rail falls by more than 5%, current rises unexpectedly, or a device becomes warm.

Fit the pull-ups and pull-downs in Section 0.3 so every signal has its specified state before firmware starts. Use temporary 1 kOhm series resistors when first connecting two potentially driven nodes. Record idle current after every phase. Unless stated otherwise, keep all chips from later phases out of their sockets.

8.1 Phase 0 - Empty Sockets and Power Distribution

Install: Breadboards, sockets, decoupling capacitors, pull-ups, power wiring, and signal wiring. Install no active device, including the Pico 2.

Implementation: [Phase 0 power checklist](#).

Test plan:

1. With every IC still removed, verify each socket's occupied rows, notch direction, pin-1 corner, and width against Section 3.1. Mark pin 1 on the breadboard and socket with a paint pen, and photograph the empty-board orientation before wiring over the socket outlines.
2. With power disconnected, check resistance from each supply rail to ground. Investigate readings below 1 kOhm after capacitors charge.
3. **GAL removed:** Check every address, data, and control net end-to-end, then verify no continuity between neighboring bus lines. At the empty GAL socket, verify each signal has no unintended short to GND, 5 V, or an adjacent pin. Do not attempt to verify GAL output levels while the GAL is removed.
4. Apply 5 V and measure every 5 V-powered DIP socket supply pin. Require 4.75 V to 5.25 V at VCC and less than 50 mV at each ground pin. The AHCT245 and GAL VCC socket contacts must read 5 V, while the LVC245 and LVC244 VCC contacts must remain at 0 V because their 3.3 V source, the absent Pico, is not yet installed. With the GAL removed, do not test GAL output levels. Verify the GAL socket and associated nets have no unintended continuity to GND, 5 V, or adjacent signals. GAL output-level verification is performed in Phase 2 after the GAL is installed.
5. If using the photographed plug-in supply, confirm that its body obscures no more than Core Board rows 1-3. Load its 5 V output to at least 500 mA, require 4.75 V to 5.25 V at the farthest board, and confirm no regulator becomes too hot to touch. Leave its 3.3 V output disconnected.
6. Verify the external +5 V rail reaches Pico VSYS only through the 1N5819 and does not reach Pico VBUS, the 3.3 V rail, or any GPIO contact. With external power applied, VSYS must be one Schottky drop below the +5 V rail. Verify each 5 V-side active-low control is pulled HIGH. With power removed, measure approximately 10 kOhm from every Pico-side pull-up contact to the unpowered 3.3 V rail and from every pull-down contact to GND, as listed in Section 0.3; powered Pico-side logic levels are checked in Phase 1. Measure approximately 10 kOhm from each GP10-GP17 contact to GND through the data SIP network.

Pass gate: No shorts or crossed nets, correct supply voltage at every socket, and negligible current with all devices removed.

8.2 Phase 1 - Raspberry Pi Pico 2 Supervisor

Install: Pico 2 only.

Firmware feature: A diagnostic image must establish safe output levels before enabling any GPIO output: GP7 and GP9 LOW to isolate the data path and hold MCP RESET# asserted; GP3 LOW to assert Z80 RESET#; GP4, GP5, GP21, GP22, and GP26 HIGH to deassert BUSREQ#, SRAM CE#, SPI CS#, SRAM WE#, and SRAM OE#; GP2 LOW to stop the clock; and GP6 LOW for the inactive data direction. GP8 remains an input. It must also provide a slow walking-one GPIO test selected through the USB serial console.

Implementation: [Phase 1 application](#), backed by the shared [supervisor module](#).

Test plan:

1. With the Section 0.4 1N5819 fitted, USB and external power may be connected together. Confirm neither source back-powers the other, then require 3.20 V to 3.40 V on the Pico 3.3 V rail, at the still-absent SN74LVC245AN and SN74LVC244AN VCC contacts.
2. Scope GP7 and GP9 through reset and startup; GP7 must remain LOW and GP9 must remain LOW so both bus interfaces stay isolated.
3. Verify GP3 and the directly connected Z80 RESET# socket pin are LOW, and all other Pico control pins have the inactive levels listed above before, during, and after startup. Verify the RESET# node never exceeds the Pico 3.3 V rail; no 5 V pull-up is permitted on it.
4. Before configuring GP10-GP17 as outputs, require all eight to read LOW from the external SIP network. Drive each HIGH in turn and verify 3.20-3.40 V while the other seven remain LOW.
5. Run the walking-one test and probe each destination socket. Require one-to-one routing, 0 V/3.3 V levels, and no change on neighboring pins. Restore safe levels when the test ends or the USB link drops.

Pass gate: Stable 3.3 V, safe startup levels, and correct routing for every Pico signal.

8.3 Phase 2 - ATF22V10 Arbitration and SN74HCT541 Buffer

Install: Program and verify the ATF22V10 outside the circuit, then install it with the SN74HCT541 and SRAM still removed. After the GAL truth-table tests pass, install the HCT541. Keep the Z80, MCP23S17, and SRAM removed. Connect GAL pins 9/10/11 to GP7/GP9/GP6 with their fitted pull-downs, tie only unused pin 13 to GND, and tie both HCT541 output-enable pins LOW. Require 4.75 V to 5.25 V at GAL VCC before testing any pulled-up GAL output.

Firmware feature: Add commands to toggle each supervisor output at 10 Hz and generate selectable 1 kHz, 100 kHz, and 1 MHz 50% duty-cycle clocks on GP2.

Implementation: [Phase 2 application](#), using the shared [clock module](#).

Test plan:

1. Require a successful programmer readback/verify of the exact `src/pld/sram_control.pld` JEDEC image before inserting the GAL.
2. With RESET# LOW, toggle Pico CE#/OE#/WE# one at a time and require the corresponding GAL pin 16/15/14 to follow while the other two remain HIGH. Then set RESET# HIGH and manually exercise pulled-up BUSACK#/MREQ#/RD#/WR# through 1 kOhm; verify all three CPU-side truth table paths and the BUSACK# LOW override.
3. With each Pico and Z80 candidate control held HIGH, toggle RESET# and BUSACK# separately while observing GAL pins 14-16 on the scope. The consensus terms must hold every output continuously HIGH; any active-low pulse fails the programmed image.
4. **GAL installed; data transceivers removed:** Hold DATA_ENABLE pin 9 LOW and toggle DATA_DIR pin 11; GAL pins 17 and 18 must both remain HIGH. Drive DATA_ENABLE HIGH: pin 17 must be LOW only when DATA_DIR is HIGH, and pin 18 must be LOW only when DATA_DIR is LOW. Scope both outputs while changing DATA_DIR with DATA_ENABLE LOW; neither may pulse LOW.
5. Install the HCT541. Use the Stage 2 walking command to toggle its eight functional input paths independently; RESET# was already tested in Phase 1 and is not an HCT541 input. At the selected output require LOW below 0.3 V, HIGH at or above 4.4 V, correct polarity, and no activity on adjacent outputs.
6. Test HCT541 channel 1 at each clock frequency. Require 45% to 55% duty cycle and clean transitions at Z80 socket pin 6.
7. After the Pico 3.3 V rail reaches 3.20 V, verify RESET# remains LOW while BUSREQ#, SRAM CE#/WE#/OE#, and SPI CS# remain HIGH for at least 100 ms. Before 3.3 V is valid, RESET# must remain LOW but SRAM control levels are not used as a retention guarantee.
8. Power-cycle ten times while monitoring these signals. Any active-low transition after 3.3 V becomes valid fails the phase.

Pass gate: Programmer verification and every GAL truth-table case pass, all eight HCT541 outputs have valid 5 V levels and correct polarity, the 1 MHz clock is clean, and startup creates no active-low glitch.

8.4 Phase 3 - MCP23S17 SPI Address Generator

Install: The 3.3 V-powered SN74LVC244AN first, then Q1 and the MCP23S17-E/SP. The already-tested HCT541 supplies all three SPI inputs. Keep Z80 and SRAM removed; their empty sockets expose the pulled-up shared address bus for probing.

Electrical hold point: Fit level translation on all SPI inputs. The MCP23S17 datasheet specifies $V_{IH} \geq 0.8V_{DD}$ for CS#, SCK, and SI, which is 4.0 V with a 5 V supply; a Pico 3.3 V HIGH is therefore not compliant. Translate Pico CS#, SCK, and MOSI through SN74HCT541 channels 3-5 (Section 4). Buffer MCP SO/MISO down through SN74LVC244AN channel 2A1/2Y1 (Section 5.3); do not connect it directly to GP20. Tie the MCP23S17 A0/A1/A2 hardware address pins to GND (Section 2). Confirm this wiring against the schematic before proceeding.

Firmware feature: Add byte-level SPI register read/write, a write-then-read register test, and 16-bit walking-one/walking-zero tests that can configure both MCP ports as either inputs or outputs.

Implementation: [Phase 3 application](#), using the shared [MCP23S17 driver](#).

Test plan:

1. Before fitting the MCP, pull each used LVC244 input LOW through 1 kOhm, then release it HIGH through its fitted 10 kOhm pull-up. Verify the matching Pico input reads LOW and HIGH at 3.3 V levels and unused channels do not change.
2. With GP9 LOW, verify Q1 holds MCP RESET# LOW and all A0-A15 nodes read HIGH through the two SIP networks. Drive GP9 HIGH and require a clean 5 V RESET# release.
3. With compliant translation fitted, write and read back 0x55 and 0xAA in IODIRA, IODIRB, OLATA, and OLATB.
4. Configure outputs and probe walking-one and walking-zero patterns at the empty Z80 and SRAM sockets.
5. Configure inputs, apply 0 V or 5 V through 10 kOhm to each pin, and verify only the corresponding GPIO register bit changes.
6. Run 10,000 alternating register writes and reads with zero errors.

Pass gate: Compliant SPI levels, error-free register access, and correct operation of all 16 port bits in both directions.

8.5 Phase 4 - Direct Address Bus and Reset Isolation

Install: No additional IC. Keep Z80 and SRAM removed.

Firmware feature: Address helpers must assert ADDR_ENABLE LOW, release MCP reset, wait, preload OLAT, and only then set IODIR outputs. On every exit they must assert reset again.

Implementation: [Phase 4 application](#), using the shared [bus module](#).

Test plan:

1. With ADDR_ENABLE LOW, verify RESET# LOW and all 16 MCP pins high-impedance; each bus line must sit HIGH through its 10 kOhm pull-up.
2. Release reset, select MCP output direction, and test 0x55, 0xAA, walking-one, and walking-zero patterns on every shared address line.
3. Assert reset and use a 1 kOhm test pull-down to prove each bus line moves independently while MCP is isolated.
4. Release reset with IODIR inputs. Drive each bus line through 1 kOhm and verify the MCP reads it without driving back.
5. Test 0x0000, 0xFFFF, 0x5555, 0xAAAA, and a walking one across A0-A15, repeat with a walking zero, then run 1,000 release/configure/reset cycles while checking current. Scope A0, A7, A8, and A15 at their SRAM socket contacts during 0x0000, 0xFFFF, 0x5555, and 0xAAAA; require valid levels without double-clocking, sustained mid-rail plateaus, or ringing that crosses the MCP/SRAM input thresholds.

Pass gate: Every address bit passes in both directions and MCP reset reliably returns every port to high-impedance input mode.

8.6 Phase 5 - All-PDIP Data Transceivers and Interlock

Install sequence: Start with the verified GAL installed and both data transceivers removed. Perform test 1, power off, then install the SN74AHCT245N and SN74LVC245AN in that order. Keep Z80 and SRAM removed. Tie AHCT DIR HIGH and LVC DIR LOW before insertion.

Firmware feature: Add an 8-bit data-bus test using the same disable-change-enable sequence and fixed, walking-one, and walking-zero patterns on the Pico data GPIOs.

Implementation: [Phase 5 application](#), using the shared [bus module](#).

Test plan:

1. **GAL installed; data transceivers removed:** With GP6/GP7 LOW, verify GAL pins 17 and 18 are both HIGH. Toggle GP7 HIGH and GP6 LOW/HIGH; require exactly one GAL OE# output to go LOW and verify the other remains HIGH.
2. **GAL and data transceivers installed:** With GP7 LOW, verify GAL pins 17/18 and both transceiver OE# pin 19 contacts are HIGH. Confirm both A and B buses remain high-impedance and neither supply current changes abnormally. This is the end-to-end safety-property check: DATA_ENABLE LOW must disable both physical data drivers. Verify the AHCT A1-A8 pins remain below 0.8 V while Pico data GPIOs are inputs.
3. Select Pico-to-bus direction and test 0x00, 0xFF, 0x55, 0xAA, walking-one, and walking-zero. Verify levels and bit order.
4. Drive GP7 LOW, change GP6 LOW, and drive GP7 HIGH. Drive each bus input with 0 V and 5 V through 1 kOhm and verify the Pico reading.
5. Run 1,000 disable-change-enable cycles while checking readback and supply current.

Pass gate: All eight bits pass both ways, isolation works, and no direction change causes contention or unexpected current.

8.7 Phase 6 - AS6C1008 SRAM and DMA Path

Install: The AS6C1008-55PCN only after the programmed ATF22V10, HCT541 channels 6-8, and final SRAM-side pull-ups have passed Phase 2. Keep the Z80 removed.

Electrical hold point: During DMA the GAL must select only Pico CE#/OE#/WE#; during execution it must select only Z80 MREQ#/RD#/WR#. CE# must not be tied LOW, and no Pico output is joined directly to a Z80 output. With the Z80 removed, BUSACK# floats HIGH via its pull-up; hold RESET# LOW so the programmed equations select the Pico side regardless. Do not install SRAM until all three GAL outputs and all three corresponding HCT541 outputs have passed static truth-table, continuity, and voltage-level tests.

Firmware feature: Add single-byte DMA read/write primitives, a walking address/data test, a two-pass full-memory pattern test, and a March C- or equivalent RAM test. Every failure must report its address, expected byte, and actual byte over USB serial.

Implementation: [Phase 6 application](#), using the shared [SRAM DMA module](#).

Test plan:

1. With transceivers disabled, require A16 LOW, CE2 HIGH, and CE#, OE#, and WE# HIGH directly at the SRAM.
2. In DMA mode, write and read one byte. Require WE# HIGH before address or data changes, and disable the Pico data driver before asserting OE# for readback.
3. Write unique values at 0x0000 and each power-of-two address from 0x0001 through 0x8000. Verify every value remains independent.
4. At several addresses test 0x00, 0xFF, 0x55, 0xAA, walking-one, and walking-zero data.
5. Fill all 65,536 bytes with the XOR of the address bytes, verify it, then repeat with the complement. Run the March test afterward.
6. Repeat after ten power cycles and with the intended 1 MHz timing.

Pass gate: Zero address, data, full-range pattern, or March-test errors and no overlap between CPU and Pico SRAM control sources.

8.8 Phase 7 - Z84C0020PEC CPU, Installed Last

Install: Z84C0020PEC. Preload SRAM first. Disable both bus transceivers, drive the Pico-side SRAM CE#/OE#/WE# controls inactive, hold RESET# LOW and BUSREQ# HIGH, and stop the clock LOW before insertion and power-up. RESET# LOW deliberately keeps the programmed logic on the inactive Pico side; the GAL equations switch to CPU controls automatically only when RESET# is released with BUSACK# HIGH.

Firmware feature: Add clock single-step and selectable 10 Hz, 1 kHz, 100 kHz, and 1 MHz run modes; reset pulse control; timed BUSREQ#/BUSACK# acquisition; and a command that preloads and verifies a small test program before releasing reset.

Implementation: [Phase 7 application](#), using the shared [CPU ownership module](#).

Test plan:

1. Verify RESET# LOW, BUSREQ# HIGH, both transceiver OE# signals HIGH, and SRAM write inactive immediately after power-up.
2. Clock at 10 Hz with RESET# asserted for at least three full cycles. Release reset and verify the first opcode fetch at 0x0000, including the expected M1#, MREQ#, and RD# sequence.
3. Preload NOP; NOP; JP 0000h. Single-step and verify address progression and the repeating jump while Pico bus drivers stay high-impedance.
4. Repeat at 1 kHz, 100 kHz, and 1 MHz with no unexpected SRAM writes.
5. Assert BUSREQ# while clocking. Require BUSACK# LOW after the current machine cycle and verify CPU bus outputs are high-impedance. Release BUSREQ# and require execution to resume.
6. Assert RESET# for at least three clocks during execution, release it, and verify a new fetch from 0x0000.

Pass gate: Correct reset fetch and execution through 1 MHz, valid BUSREQ#/BUSACK# transfer, and no bus contention.

8.9 Phase 8 - Integrated Virtual-ROM Boot and I/O Trap

Install: No further chips. Load final supervisor firmware.

Firmware feature: Combine safe startup, timed bus acquisition, image injection and readback, run control, and the synchronous IN/OUT trap. Maintain counters for boots, DMA failures, readback mismatches, trap timeouts, and unexpected RD#/WR# control states.

Implementation: [Phase 8 application](#), using the shared [I/O trap](#), [CPU](#), and [SRAM](#) modules.

Test plan:

1. Acquire ownership by one of two complete procedures. Either use the timed BUSREQ#/BUSACK# handshake, or drive the Pico SRAM controls inactive, assert RESET#, continue the clock for at least three full cycles, and then stop it. In either case, verify the Z80 address and data buses are high-impedance before enabling a transceiver. The Section 1.2 AND gate then grants the Pico exclusive SRAM control; inject a known 64 KB image, read back every byte, and compare it before CPU release.
2. Run a RAM increment loop for one hour at 1 MHz without corruption or supervisor timeout.
3. Execute OUT on ports 0x00, 0x01, 0x55, 0xAA, and 0xFF with matching data. Verify correct clock stop, address/data capture, driver disable, and clock resume.
4. Execute matching IN tests and verify each Pico-generated byte is stored correctly in SRAM.
5. With a logic analyzer, prove both transceiver OE# signals are HIGH during CPU cycles, the CPU bus is high-impedance before DMA enable, both DIR controls change only while their OE# is HIGH, and no overlap occurs between SRAM control sources.
6. Repeat cold boot, injection, execution, and I/O tests 100 times with zero logged failures.

Pass gate: Zero image or boot failures, correct IN/OUT behavior, one-hour stable execution, and contention-free ownership transitions.

8.10 Phase 9 - Flash Disk Image Loader and CP/M Storage

Install: No hardware rework. Set `PICO_FLASH_SIZE_BYTES` to `0x290000` in `CMakeLists.txt`, define `PICO_FLASH_ASSUME_CORE1_SAFE=1`, and link the libraries listed in Section 6.3. Provision the manifest-backed boot package and all four 320 KiB disk slots with the verified `picotool` commands there.

Firmware feature: With `RESET#` held LOW, recover any valid journal, validate the boot manifest and CRC32, DMA-write its payload to SRAM, and compare every byte before `RESET#` release. Do not wait for `BUSACK#` during this cold-boot path: `RESET#` itself selects the Pico's SRAM controls through Section 1.2's GAL equations. Once running, ports `0x10-0x14` provide command/status, drive, 16-bit LBA, and 128-byte data transfers. Reads are synchronous XIP copies; writes use the journaled core-1 service and `BUSY/READY/ERROR` status defined in Section 8.13.

Implementation: [Phase 9 application](#), with the shared [disk device](#), [flash backend](#), and [flash layout](#).

Test plan:

1. With the Z80 socket populated and Phase 8 passing, confirm `io_trap_handler()` still passes every Phase 8 IN/OUT test unchanged; Phase 9 adds no new pins or rework to disturb it.
2. Confirm the link fails if code crosses `0x10290000`, while a runtime print/static assertion still reports the physical C macro as 4 MiB.
3. Provision the boot package and four exact-size disk images with `picotool -v`; read every region's first and last page back and compare host CRC32 values.
4. Cold-boot with `RESET#` held LOW. Require journal recovery and SRAM verification to finish without waiting for `BUSACK#`, then release `RESET#` only after success. Corrupt the manifest, payload, and SRAM readback separately and require each case to remain fail-closed.
5. Exercise all 2,560 LBAs on every drive through ports `0x10-0x14`. Verify exact 128-byte transfers, invalid drive/LBA rejection, command while `BUSY` rejection, and test-injected queue-full/error clearing behavior. After an injected flash or journal failure, require every `READ/WRITE` command to retain `READYIERROR` until reboot recovery.
6. Probe `BUSREQ#`, `BUSACK#`, `IORQ#`, and `CLK` during a write. Require the trap to remain armed until `BUSACK#` is LOW and to be armed again before `BUSACK#` returns HIGH, with no untrapped I/O edge in either interval. Inject an `IORQ#` falling edge while `BUSACK#` is LOW and require the handler to disable the IRQ without touching `CLK`, `SPI0`, or either bus.
7. Add test-only power-cut hooks after journal-data program, header program, target erase, partial target program, target verification, and header clear. Reboot after every hook and require recovery to produce either the complete old block (before valid header) or the complete new block (after valid header), never a mixture.
8. Repeat reads and writes with Wi-Fi absent, associating, connected, and reconnecting. Disk completion must not depend on network state, and WebSocket queue overflow must remain counted rather than block.
9. Rewrite hot directory blocks repeatedly while tracking journal-pair rotation and the flash part's rated erase endurance. Treat this as a smoke test, not proof of lifetime.
10. Inject safe-execute entry and exit failures. Require `RESET#` LOW, isolated buses, stopped `CLK`, and a watchdog reboot without waiting on the core-0 release queue; recovery must retain the verified old or new disk block.

Pass gate: Boot and all four disks match host CRC32 values, every fault-injection reboot recovers an intact old or new block, all bounds and manifest failures remain fail-closed, disk service works without Wi-Fi, the linker protects the storage boundary, and logic-analyzer captures show no untrapped Z80 cycle around a flash write.

8.11 Phase 10 - WebSocket Terminal Console

Install: No further bus hardware. Use a Pico 2 W for the final networked terminal build, or keep the same firmware hooks compiled as stubs on a non-W Pico 2.

Firmware feature: Start the WebSocket console service on core 1 after core 0 has completed safe GPIO startup, queue initialization, and the Phase 9 boot-image load (which finishes entirely on core 0 before core 1 is launched). Core 0 continues to own the Z80 clock, bus transceivers, MCP23S17, SRAM DMA, and I/O trap. Core 1 owns Wi-Fi connection management, the embedded HTTP terminal page, WebSocket client state, network polling, and the Section 6.3 flash disk-write service -- all in the same `core1_main()` task, since `multicore_launch_core1()` only accepts one entry point. The two cores exchange only terminal bytes, flash disk-write requests, and status through nonblocking queues, following the `pico-altair-8800` console bridge pattern.

Implementation: [Phase 10 application](#), with the shared [terminal bridge](#), [network service](#), and [browser terminal](#).

Test plan:

1. Boot with no browser connected. Verify the Z80 still runs the Phase 8 I/O tests, IN 0x01 reports no client, and terminal output does not accumulate without bound.
2. Connect a browser to <http://<pico-ip>:8088/>, or a WebSocket client to <ws://<pico-ip>:8088/>. Verify IN 0x01 sets the client-connected bit without disturbing the Z80 clock.
3. Run a Z80 program that writes a continuous alphabet pattern to OUT 0x00. Verify the browser receives the stream in order and that queue-full conditions are counted rather than blocking the trap.
4. Type from the browser and verify the Z80 receives each byte through IN 0x00 only after IN 0x01 reports data available.
5. Disconnect and reconnect the browser while the Z80 test program runs. Verify stale input is cleared, output resumes for the new client, and no trap timeout counter increments.
6. Exhaust the default alarm pool before service startup and require the supervisor to remain fail-closed rather than launching core 1 without both WebSocket polling timers.

Pass gate: The WebSocket service remains responsive while the Z80 runs at the Phase 8 qualified 1 MHz setting, no network path runs on the core that services Z80 timing, and all terminal queue overflow or client disconnect conditions are visible through counters rather than blocking the CPU trap.

8.12 Frequency Qualification

Begin only after Phase 10 passes at 1 MHz. Test 2 MHz, then increase in 500 kHz steps to 4 MHz. At each step repeat SRAM readback, the one-hour memory loop, and continuous IN/OUT tests while measuring stop latency. At 1, 2, 3, and 4 MHz also apply address patterns 0x0000, 0xFFFF, 0x5555, 0xAAAA, walking one, and walking zero while capturing A0, A7, A8, and A15 at the SRAM pins. Capture CLK, MREQ#, RD#/WR#, SRAM CE#/OE#/WE#, and D0-D7 as well; require valid read data before the Z80 setup deadline and every SRAM write pulse to meet the 45 ns minimum after propagation through the GAL and HCT541. The qualified frequency is the highest error-free step for which the logic analyzer proves the clock always stops before the Z80 advances beyond the safe trap point. Do not claim operation above 4 MHz without equivalent timing evidence.

8.13 Pico Diagnostic Firmware Core Features

These Pico SDK fragments show the safety-critical core of each test and finish with the required two-core `main()` integration order. Board-specific Wi-Fi/WebSocket hooks and the optional nonblocking USB command parser remain external. Every diagnostic command must print PASS or a detailed failure and call `isolate_buses()` before returning. PIN_BUSACK_N is GP0, driven from Z80 BUSACK# (pin 23) through the SN74LVC244AN input buffer (Section 5.3); IORQ#, RD#, and MCP SO use the same buffer, so no 5 V output reaches the Pico directly. PIN_SRAM_CE_N is fixed at GP5 (ATF22V10 pin 7; Section 1.2). GP23 on the Pico 2 W is dedicated to the CYW43439 wireless module's control interface (shared with GP24/GP25/GP29) and must never be repurposed. PIN_DATA_0- PIN_DATA_7 occupy GP10-GP17. There is no PIN_SRAM_SOURCE_SELECT GPIO: the ATF22V10 combines RESET# and BUSACK# inside each programmed SRAM-control equation (Section 1.2), so ownership still switches automatically with no extra Pico pin. PIN_WR_N is GP28, buffered through the same SN74LVC244AN (Section 5.3); `io_trap_handler()` reads both PIN_RD_N and PIN_WR_N to resolve cycle intent.

Safe Startup and Walking Output (Phases 1-2)

Maintained source: [pins.h](#), [supervisor.h](#), and [supervisor.c](#).

Preload each output latch while the pin is still an input, then enable the output driver. This prevents a brief LOW pulse on active-low lines.

```
#include <stdint.h>
#include <stdio.h>
#include "pico/stdlib.h"

enum {
    PIN_IORQ_N = 1, PIN_CLK = 2, PIN_RESET_N = 3,
    PIN_BUSREQ_N = 4, PIN_BUSACK_N = 0,
    PIN_DATA_DIR = 6, PIN_DATA_ENABLE = 7,
```

```

PIN_UNUSED_8 = 8, PIN_ADDR_ENABLE = 9,
PIN_DATA_0 = 10, PIN_DATA_1 = 11, PIN_DATA_2 = 12, PIN_DATA_3 = 13,
PIN_DATA_4 = 14, PIN_DATA_5 = 15, PIN_DATA_6 = 16, PIN_DATA_7 = 17,
PIN_SPI_SCK = 18, PIN_SPI_MOSI = 19, PIN_SPI_MISO = 20,
PIN_SPI_CS_N = 21,
PIN_SRAM_WE_N = 22, PIN_SRAM_CE_N = 5, PIN_SRAM_OE_N = 26,
PIN_RD_N = 27, PIN_WR_N = 28
};

static void output_with_initial_level(uint pin, bool level) {
    gpio_init(pin);
    gpio_put(pin, level);      // Preload SIO output latch.
    gpio_set_dir(pin, GPIO_OUT);
}

static void input_with_no_pull(uint pin) {
    gpio_init(pin);
    gpio_set_dir(pin, GPIO_IN);
    gpio_disable_pulls(pin);
}

static void diagnostic_safe_startup(void) {
    output_with_initial_level(PIN_DATA_ENABLE, 0);
    output_with_initial_level(PIN_ADDR_ENABLE, 0);
    output_with_initial_level(PIN_RESET_N, 0); // Hold CPU reset.
    output_with_initial_level(PIN_BUSREQ_N, 1);
    output_with_initial_level(PIN_SRAM_WE_N, 1);
    output_with_initial_level(PIN_SRAM_CE_N, 1);
    output_with_initial_level(PIN_SRAM_OE_N, 1);
    output_with_initial_level(PIN_SPI_CS_N, 1);
    output_with_initial_level(PIN_CLK, 0);
    output_with_initial_level(PIN_DATA_DIR, 0);
    input_with_no_pull(PIN_BUSACK_N);
    input_with_no_pull(PIN_IORQ_N); // Section 0.3 pulls up the LVC244 input.
    input_with_no_pull(PIN_RD_N);
    input_with_no_pull(PIN_WR_N);
}

static void walking_output_test(const uint *pins, size_t count) {
    for (size_t active = 0; active < count; ++active) {
        for (size_t index = 0; index < count; ++index)
            gpio_put(pins[index], index == active);
        sleep_ms(250);      // Probe or logic-analyzer capture point.
    }
    diagnostic_safe_startup();
}

```

Run `walking_output_test()` only in Phases 1-2 while the destination driver/bus chips are absent. It deliberately changes raw pin levels and is not safe as an in-system diagnostic after Phase 3.

Variable-Frequency Clock Generation (Phase 2, Phases 7-8 Run Modes)

Maintained source: [clock.h](#) and [clock.c](#).

Configure `PIN_CLK` as a PWM output once during Phase 2 bring-up. Reuse the same slice for the selectable 10 Hz/1 kHz/100 kHz/1 MHz run modes in Phases 7-8, and to freeze the clock during the Phase 8 I/O trap.

```

#include "hardware/pwm.h"
#include "hardware/clocks.h"

static bool set_z80_clock_hz(uint32_t hz) {
    if (hz < 10 || hz > 4000000)
        return false;

    uint slice_num = pwm_gpio_to_slice_num(PIN_CLK);
    pwm_set_enabled(slice_num, false);
    uint32_t sys_clk = clock_get_hz(clk_sys);
    uint64_t sys_clk16 = (uint64_t)sys_clk * 16u;
    uint32_t best_divider16 = 0;
    uint32_t best_count = 0;
    uint64_t best_error = UINT64_MAX;

    // Search all legal 8.4 dividers; this runs only when frequency changes.
    for (uint32_t divider16 = 16; divider16 <= 4095; ++divider16) {
        uint64_t denominator = (uint64_t)hz * divider16;
        uint64_t count = (sys_clk16 + denominator / 2u) / denominator;
    }
}

```

```

if (count < 1) count = 1;
if (count > 65536) count = 65536;
uint64_t product = (uint64_t)divider16 * count;
uint64_t target_product = (uint64_t)hz * product;
uint64_t error = sys_clk16 > target_product ?
    sys_clk16 - target_product : target_product - sys_clk16;
uint64_t best_product = (uint64_t)best_divider16 * best_count;
if (best_divider16 == 0 ||
    error * best_product < best_error * product) {
    best_divider16 = divider16;
    best_count = (uint32_t)count;
    best_error = error;
}
}

uint16_t wrap = (uint16_t)(best_count - 1u);
pwm_set_clkdiv(slice_num, (float)best_divider16 / 16.0f);
pwm_set_wrap(slice_num, wrap);
pwm_set_chan_level(slice_num, PWM_CHAN_A,
    (uint16_t)(best_count / 2u)); // Exact 50% when count is even.
pwm_set_counter(slice_num, 0);
gpio_set_function(PIN_CLK, GPIO_FUNC_PWM);
pwm_set_enabled(slice_num, true);
return true;
}

static void stop_z80_clock(void) {
    pwm_set_enabled(pwm_gpio_to_slice_num(PIN_CLK), false); // May freeze HIGH or LOW.
}

static void resume_z80_clock(void) {
    pwm_set_enabled(pwm_gpio_to_slice_num(PIN_CLK), true);
}

```

MCP23S17 Register and Port Test (Phases 3-4)

Maintained source: [mcp23s17.h](#) and [mcp23s17.c](#).

The SPI translator sits between these Pico pins and the 5 V MCP23S17; MISO/SO returns through the SN74LVC244AN. The register test proves communication before either address transceiver is enabled.

```

#include "hardware/spi.h"

enum { MCP_WRITE = 0x40, MCP_READ = 0x41 };
enum { IODIRA = 0x00, IODIRB = 0x01, GPIOA = 0x12, GPIOB = 0x13,
    OLATA = 0x14, OLATB = 0x15 };

static void mcp_spi_init(void) {
    output_with_initial_level(PIN_SPI_CS_N, 1);
    spi_init(spi0, 4000000);
    gpio_set_function(PIN_SPI_SCK, GPIO_FUNC_SPI);
    gpio_set_function(PIN_SPI_MOSI, GPIO_FUNC_SPI);
    gpio_set_function(PIN_SPI_MISO, GPIO_FUNC_SPI);
}

static void mcp_write(uint8_t reg, uint8_t value) {
    uint8_t frame[] = { MCP_WRITE, reg, value };
    gpio_put(PIN_SPI_CS_N, 0);
    spi_write_blocking(spi0, frame, sizeof frame);
    gpio_put(PIN_SPI_CS_N, 1);
}

static uint8_t mcp_read(uint8_t reg) {
    uint8_t tx[] = { MCP_READ, reg, 0x00 };
    uint8_t rx[sizeof tx];
    gpio_put(PIN_SPI_CS_N, 0);
    spi_write_read_blocking(spi0, tx, rx, sizeof tx);
    gpio_put(PIN_SPI_CS_N, 1);
    return rx[2];
}

static bool mcp_register_test(void) {
    const uint8_t patterns[] = { 0x55, 0xAA };
    mcp_write(IODIRA, 0x00);
    mcp_write(IODIRB, 0x00);
    for (size_t i = 0; i < sizeof patterns; ++i) {
        mcp_write(OLATA, patterns[i]);
    }
}

```



```

mcp_write(OLATB, (uint8_t)~patterns[i]);
if (mcp_read(OLATA) != patterns[i] ||
    mcp_read(OLATB) != (uint8_t)~patterns[i])
    return false;
}
return true;
}

```

Contention-Safe Bus Isolation (Phases 4-6)

Maintained source: [bus.h](#) and [bus.c](#).

ADDR_ENABLE and DATA_ENABLE must both be LOW for isolation. DATA_ENABLE must remain LOW before DATA_DIR changes.

```

static void isolate_buses(void) {
    gpio_put(PIN_ADDR_ENABLE, 0);
    gpio_put(PIN_DATA_ENABLE, 0);
}

```

Data and Address Bus GPIO Helpers (Phases 4-6)

Maintained source: [bus.h](#) and [bus.c](#).

PIN_DATA_0-PIN_DATA_7 (GP10-GP17) connect to both fixed-direction data transceivers. PIN_DATA_DIR and PIN_DATA_ENABLE feed the GAL, which enables exactly one selected path. Address bytes reach the shared bus through the MCP23S17's own output latches, already exercised in the Phase 3-4 register test. Call these helpers only while RESET# is asserted or request_cpu_bus() has returned true, and keep SRAM CE# HIGH while changing address or data direction.

```

static const uint DATA_PINS[8] = {
    PIN_DATA_0, PIN_DATA_1, PIN_DATA_2, PIN_DATA_3,
    PIN_DATA_4, PIN_DATA_5, PIN_DATA_6, PIN_DATA_7
};

static void data_bus_drive(uint8_t value) {
    gpio_put(PIN_DATA_ENABLE, 0);
    for (size_t i = 0; i < 8; ++i) {
        gpio_put(DATA_PINS[i], (value >> i) & 1); // Preload before enabling output.
        gpio_set_dir(DATA_PINS[i], GPIO_OUT);
    }
    gpio_put(PIN_DATA_DIR, 1); // Select AHCT Pico-to-bus path.
    busy_wait_us_32(1);
    gpio_put(PIN_DATA_ENABLE, 1);
}

static void data_bus_prepare_input(void) {
    gpio_put(PIN_DATA_ENABLE, 0);
    for (size_t i = 0; i < 8; ++i)
        gpio_set_dir(DATA_PINS[i], GPIO_IN);
    gpio_put(PIN_DATA_DIR, 0); // Select LVC bus-to-Pico path.
    busy_wait_us_32(1);
    gpio_put(PIN_DATA_ENABLE, 1);
}

static uint8_t data_bus_sample(void) {
    uint8_t value = 0;
    for (size_t i = 0; i < 8; ++i)
        value |= gpio_get(DATA_PINS[i]) << i;
    return value;
}

static void address_bus_drive(uint16_t address) {
    gpio_put(PIN_ADDR_ENABLE, 0);
    busy_wait_us_32(1); // MCP RESET# low-pulse minimum.
    gpio_put(PIN_ADDR_ENABLE, 1); // Release MCP reset; ports default to inputs.
    busy_wait_us_32(1);
    mcp_write(OLATA, (uint8_t)address);
    mcp_write(OLATB, (uint8_t)(address >> 8));
    mcp_write(IODIRA, 0x00);
    mcp_write(IODIRB, 0x00);
}

```

SRAM DMA Access and Pattern Test (Phase 6)

Maintained source: [sram.h](#) and [sram.c](#).

The address and data helper functions above represent the already tested MCP23S17 and GPIO bus operations. The control-source selector must grant exclusive SRAM control to the Pico before this code runs.

```
static void dma_write_byte(uint16_t address, uint8_t value) {
    gpio_put(PIN_SRAM_CE_N, 1);
    gpio_put(PIN_SRAM_OE_N, 1);
    gpio_put(PIN_SRAM_WE_N, 1);
    address_bus_drive(address);      // MCP directly drives A0-A15.
    data_bus_drive(value);          // Pico -> AHCT245 -> D0-D7.
    gpio_put(PIN_SRAM_CE_N, 0);
    busy_wait_us_32(1);
    gpio_put(PIN_SRAM_WE_N, 0);
    busy_wait_us_32(1);
    gpio_put(PIN_SRAM_WE_N, 1);
    gpio_put(PIN_SRAM_CE_N, 1);
}

static uint8_t dma_read_byte(uint16_t address) {
    uint8_t value;
    gpio_put(PIN_SRAM_CE_N, 1);
    gpio_put(PIN_SRAM_WE_N, 1);
    address_bus_drive(address);
    data_bus_prepare_input();        // Disable, reverse, then enable.
    gpio_put(PIN_SRAM_CE_N, 0);
    gpio_put(PIN_SRAM_OE_N, 0);
    busy_wait_us_32(1);
    value = data_bus_sample();
    gpio_put(PIN_SRAM_OE_N, 1);
    gpio_put(PIN_SRAM_CE_N, 1);
    gpio_put(PIN_DATA_ENABLE, 0);
    return value;
}

static bool sram_pattern_test(bool complement) {
    for (uint32_t address = 0; address < 65536; ++address) {
        uint8_t expected = (uint8_t)address ^ (uint8_t)(address >> 8);
        dma_write_byte((uint16_t)address,
            complement ? (uint8_t)~expected : expected);
    }
    for (uint32_t address = 0; address < 65536; ++address) {
        uint8_t expected = (uint8_t)address ^ (uint8_t)(address >> 8);
        if (complement)
            expected = (uint8_t)~expected;
        uint8_t actual = dma_read_byte((uint16_t)address);
        if (actual != expected) {
            printf("FAIL %04lx expected=%02x actual=%02x\n",
                (unsigned long)address, expected, actual);
            isolate_buses();
            return false;
        }
    }
    isolate_buses();
    return true;
}
```

Z80 Single-Step and Timed Bus Request (Phases 7-8)

Maintained source: [cpu.h](#) and [cpu.c](#).

Single-step uses SIO rather than PWM. `request_cpu_bus()` and `release_cpu_bus()` each take an explicit `timeout_us` bound so a wiring fault or a missing Z80 cannot hang the supervisor waiting on `BUSACK#`. The clock must be running while requesting or releasing `BUSREQ#`; after using `clock_one_cycle()`, call `set_z80_clock_hz()` to restore the CLK pin's PWM function before either handshake.

```
static void clock_one_cycle(uint32_t half_period_us) {
    pwm_set_enabled(pwm_gpio_to_slice_num(PIN_CLK), false);
    gpio_set_function(PIN_CLK, GPIO_FUNC_SIO);
    gpio_set_dir(PIN_CLK, GPIO_OUT);
    gpio_put(PIN_CLK, 0);
}
```

```

    busy_wait_us_32(half_period_us);
    gpio_put(PIN_CLK, 1);
    busy_wait_us_32(half_period_us);
    gpio_put(PIN_CLK, 0);
}

static bool request_cpu_bus(uint32_t timeout_us) {
    absolute_time_t deadline = make_timeout_time_us(timeout_us);
    isolate_buses();
    gpio_put(PIN_BUSREQ_N, 0);
    while (gpio_get(PIN_BUSACK_N) != 0) {
        if (time_reached(deadline)) {
            gpio_put(PIN_BUSREQ_N, 1);
            return false;
        }
        tight_loop_contents();
    }
    return true;           // RESET# HIGH: BUSACK# LOW selects DMA controls.
}

static bool release_cpu_bus(uint32_t timeout_us) {
    absolute_time_t deadline = make_timeout_time_us(timeout_us);
    gpio_put(PIN_SRAM_CE_N, 1);
    gpio_put(PIN_SRAM_OE_N, 1);
    gpio_put(PIN_SRAM_WE_N, 1);
    isolate_buses();
    gpio_put(PIN_BUSREQ_N, 1);
    while (gpio_get(PIN_BUSACK_N) == 0) {
        if (time_reached(deadline)) {
            gpio_put(PIN_RESET_N, 0);
            return false;
        }
        tight_loop_contents();
    }
    return true;           // BUSACK# HIGH restores CPU controls.
}

```

Synchronous I/O Trap Handler (Phase 8)

Maintained source: [io_trap.h](#) and [io_trap.c](#).

A falling-edge IRQ on `PIN_IORQ_N` freezes the clock, reverses the address transceiver with the same contention-safe helper used elsewhere, reads the trapped port from the lower MCP23S17 port (Section 6.4's 8-bit decode limit), and reuses the already-tested data bus helpers from the SRAM DMA code to sample or drive the data byte. The handler first confirms `PIN_BUSACK_N` is still HIGH; `IORQ#` tri-states along with `RD#/WR#` whenever `BUSACK#` is asserted (Section 6.3), so a falling edge seen while the Pico already owns the bus cannot be a real Z80 cycle and is ignored before any bus or SPI0 state is touched. Both MCP ports must be forced to inputs before the two transceivers' shared `OE#` is enabled, even though only GPIOA is read; otherwise the still-output GPB port fights Z80 A8-A15. For IN, the data byte must stay driven until the Z80 samples it, so the clock resumes and `RD#` is polled before the data bus is isolated.

`process_virtual_io_read` and `process_virtual_io_write` are the only application-supplied hooks. They must use the signatures below, finish without sleeping, printing, waiting on USB, or taking a lock held by main code, and must not start another bus operation. Move expensive work to the main loop through a lock-free queue.

```

#include "hardware/watchdog.h"

uint8_t process_virtual_io_read(uint8_t port);
void process_virtual_io_write(uint8_t port, uint8_t value);

enum { TRAP_RELEASE_TIMEOUT_US = 500000 }; // Covers the 10 Hz test mode.
static volatile uint32_t trap_timeout_count;
static volatile uint32_t unexpected_control_count;

static _Noreturn void reset_after_trap_fault(void) {
    isolate_buses();
    gpio_put(PIN_RESET_N, 0);
    stop_z80_clock();
    for (unsigned int cycle = 0; cycle < 3; ++cycle)
        clock_one_cycle(1); // RESET# setup and each half-cycle exceed Z80 minima.
    watchdog_reboot(0, 0, 0); // Fail-closed: same recovery path as flash faults.
    while (true)
        tight_loop_contents(); // watchdog_reboot() takes effect asynchronously.
}

```

```

static void io_trap_handler(uint gpio, uint32_t events) {
    (void)events;
    if (gpio != PIN_IORQ_N)
        return; // Only IORQ# is ever armed, but never trust a shared callback.
    if (!gpio_get(PIN_BUSACK_N)) {
        gpio_set_irq_enabled(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL, false);
        return; // The foreground release path re-arms this interrupt.
    }
    stop_z80_clock();
    gpio_put(PIN_ADDR_ENABLE, 1); // Release MCP reset.
    busy_wait_us_32(1);
    mcp_write(IODIRA, 0xFF);
    mcp_write(IODIRB, 0xFF);
    uint8_t port = mcp_read(GPIOA); // 8-bit I/O decode.
    gpio_put(PIN_ADDR_ENABLE, 0); // Reset makes all ports inputs.

    bool is_read = !gpio_get(PIN_RD_N);
    bool is_write = !gpio_get(PIN_WR_N);
    if (is_read == is_write) { // Neither or both asserted: hardware fault.
        ++unexpected_control_count;
        is_write = !is_read; // Best-effort fallback so the trap still resolves.
    }

    if (is_write) {
        data_bus_prepare_input(); // Bus -> Pico.
        uint8_t value = data_bus_sample();
        isolate_buses();
        process_virtual_io_write(port, value);
        resume_z80_clock();
    } else {
        data_bus_drive(process_virtual_io_read(port)); // Pico -> Bus.
        absolute_time_t deadline = make_timeout_time_us(TRAP_RELEASE_TIMEOUT_US);
        resume_z80_clock();
        while (!gpio_get(PIN_RD_N)) { // Hold data through the sample edge.
            if (time_reached(deadline)) {
                ++trap_timeout_count;
                reset_after_trap_fault();
                return;
            }
            tight_loop_contents();
        }
        isolate_buses();
    }
}

static void enable_io_trap(void) {
    input_with_no_pull(PIN_IORQ_N); // Pull-up is on the LVC244's 5 V input.
    gpio_acknowledge_irq(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL);
    gpio_set_irq_enabled_with_callback(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL,
        true, &io_trap_handler);
}

static void disable_io_trap(void) {
    gpio_set_irq_enabled(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL, false);
    gpio_acknowledge_irq(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL);
}

```

Flash Disk Image Loader (Final Phase 9 Integration)

Maintained source: [flash_disk.h](#), [flash_layout.h](#), [disk_device.c](#), and [flash_backend.c](#).

Loading a boot image is now a synchronous, core-0-only operation: a flash read is an ordinary memory access through the XIP-mapped pointer (Section 6.3), so no filesystem, blocking I/O call, or core 1 task is needed to bring the initial image into SRAM, unlike the SD design this replaces. Only CP/M's live disk-sector *writes* still need to run on core 1 and cross back to core 0's foreground loop, because only they need to freeze the Z80 around a flash erase/program cycle (Section 6.3).

```

#include <stdint.h>
#include <stdbool.h>
#include <stddef.h>
#include <stdio.h>
#include <string.h>

#include "hardware/flash.h"
#include "hardware/watchdog.h"
#include "pico/flash.h"
#include "pico/util/queue.h"

```

```

enum {
    FLASH_JOURNAL_BASE_OFFSET = 0x290000u,
    FLASH_JOURNAL_BYTES = 0x10000u,
    FLASH_JOURNAL_PAIR_BYTES = 2u * FLASH_SECTOR_SIZE,
    FLASH_JOURNAL_PAIR_COUNT = FLASH_JOURNAL_BYTES / FLASH_JOURNAL_PAIR_BYTES,
    FLASH_BOOT_BASE_OFFSET = 0x2A0000u,
    FLASH_BOOT_REGION_BYTES = 0x20000u,
    FLASH_BOOT_PAYLOAD_OFFSET = FLASH_BOOT_BASE_OFFSET + FLASH_SECTOR_SIZE,
    FLASH_DISK_BASE_OFFSET = 0x2C0000u,    // Section 6.3 partition table.
    FLASH_DISK_SLOT_BYTES = 0x50000u,      // 320 KiB per drive.
    FLASH_DISK_SLOT_COUNT = 4u,
    FLASH_DISK_RECORD_BYTES = 128u,
    FLASH_DISK_RECORD_COUNT = 2560u,
    SRAM_SIZE_BYTES = 65536
};

_Static_assert(PICO_FLASH_SIZE_BYTES == 4u * 1024u * 1024u,
    "Pico 2 W physical flash size changed");
_Static_assert(FLASH_DISK_SLOT_BYTES ==
    FLASH_DISK_RECORD_BYTES * FLASH_DISK_RECORD_COUNT,
    "disk geometry does not fill its slot");
_Static_assert(FLASH_BOOT_PAYLOAD_OFFSET + SRAM_SIZE_BYTES <=
    FLASH_BOOT_BASE_OFFSET + FLASH_BOOT_REGION_BYTES,
    "boot payload exceeds its reserved region");
_Static_assert(FLASH_DISK_SLOT_COUNT * FLASH_DISK_SLOT_BYTES == PICO_FLASH_SIZE_BYTES,
    "disk slots must end at physical flash boundary");
_Static_assert(FLASH_JOURNAL_PAIR_COUNT == 8,
    "journal layout no longer matches the partition table");
_Static_assert(PICO_FLASH_ASSUME_CORE1_SAFE,
    "core 0 journal recovery runs only before core 1 is launched");

static const uint8_t *flash_disk_slot_ptr(unsigned drive) {
    uint32_t offset = FLASH_DISK_BASE_OFFSET + drive * FLASH_DISK_SLOT_BYTES;
    return (const uint8_t *) (XIP_BASE + offset);
}

static uint32_t crc32_bytes(const uint8_t *data, size_t length) {
    uint32_t crc = UINT32_MAX;
    for (size_t i = 0; i < length; ++i) {
        crc ^= data[i];
        for (unsigned int bit = 0; bit < 8; ++bit)
            crc = (crc >> 1) ^ (0xEDB88320u & (uint32_t)~(int32_t)(crc & 1u));
    }
    return ~crc;
}

enum { Z80_BOOT_MAGIC = 0x5442385Au, Z80_BOOT_VERSION = 1u }; // "Z8BT".

typedef struct {
    uint32_t magic;
    uint16_t version;
    uint16_t header_bytes;
    uint32_t image_bytes;
    uint32_t image_crc32;
    uint32_t header_crc32;
} z80_boot_manifest_t;

_Static_assert(sizeof(z80_boot_manifest_t) == 20,
    "boot manifest layout must match the host packer");

// Core 0 only, entirely synchronous: a flash read needs no filesystem,
// no queue, and no core 1 task, unlike the SD design this replaces.
static bool prepare_reset_held_dma(void) {
    isolate_buses();
    gpio_put(PIN_RESET_N, 0);
    for (unsigned int cycle = 0; cycle < 3; ++cycle)
        clock_one_cycle(1); // Bit-banged SIO pulses; set_z80_clock_hz() runs later.
    stop_z80_clock();

    if (!gpio_get(PIN_BUSACK_N) || !gpio_get(PIN_IORQ_N) ||
        !gpio_get(PIN_RD_N) || !gpio_get(PIN_WR_N))
        return false;

    return true; // RESET# LOW selects the Pico SRAM controls (Section 1.2).
}

static bool boot_image_from_flash(void) {
    const z80_boot_manifest_t *manifest =
        (const z80_boot_manifest_t *) (XIP_BASE + FLASH_BOOT_BASE_OFFSET);
    const uint8_t *source =
        (const uint8_t *) (XIP_BASE + FLASH_BOOT_PAYLOAD_OFFSET);

```

```

if (manifest->magic != Z80_BOOT_MAGIC ||
    manifest->version != Z80_BOOT_VERSION ||
    manifest->header_bytes != sizeof(*manifest) ||
    manifest->image_bytes == 0 || manifest->image_bytes > SRAM_SIZE_BYTES)
    return false;

uint32_t header_crc = crc32_bytes((const uint8_t *)manifest,
    offsetof(z80_boot_manifest_t, header_crc32));
if (header_crc != manifest->header_crc32 ||
    crc32_bytes(source, manifest->image_bytes) != manifest->image_crc32)
    return false;

if (!prepare_reset_held_dma())
    return false;

for (uint32_t i = 0; i < manifest->image_bytes; ++i)
    dma_write_byte((uint16_t)i, source[i]);

bool ok = true;
for (uint32_t i = 0; i < manifest->image_bytes; ++i) {
    if (dma_read_byte((uint16_t)i) != source[i]) {
        printf("boot verify failed near %04lx\n", (unsigned long)i);
        ok = false;
        break;
    }
}

if (ok)
    printf("loaded boot bytes=%lu crc32=%08lx\n",
        (unsigned long)manifest->image_bytes,
        (unsigned long)manifest->image_crc32);
isolate_buses();
// RESET# remains asserted; the caller releases it only after success.
return ok;
}

static bool flash_recover_journal(void);

static bool boot_cpm_from_flash(void) {
    return flash_recover_journal() && boot_image_from_flash();
}

enum { FLASH_SERVICE_ACQUIRE_BUS, FLASH_SERVICE_RELEASE_BUS };

typedef struct {
    uint8_t operation;
} flash_service_request_t;

static queue_t flash_service_request_queue; // Core 1 -> Core 0.
static queue_t flash_service_result_queue; // Core 0 -> Core 1.

static void flash_service_queues_init(void) {
    queue_init(&flash_service_request_queue, sizeof(flash_service_request_t), 4);
    queue_init(&flash_service_result_queue, sizeof(bool), 4);
}

// Core 0's foreground loop only; never called from io_trap_handler().
static void core0_service_flash_requests(void) {
    flash_service_request_t request;
    if (!queue_try_remove(&flash_service_request_queue, &request))
        return;

    bool ok;
    switch (request.operation) {
        case FLASH_SERVICE_ACQUIRE_BUS:
            ok = request_cpu_bus(500000);
            if (ok)
                disable_io_trap(); // BUSACK# LOW: no new cycle can start.
            break;
        case FLASH_SERVICE_RELEASE_BUS:
            enable_io_trap(); // Arm before BUSACK# lets the Z80 run.
            ok = release_cpu_bus(500000);
            if (!ok)
                disable_io_trap(); // release_cpu_bus() asserted RESET#.
            break;
        default:
            ok = false;
            break;
    }
    queue_add_blocking(&flash_service_result_queue, &ok);
}

```

```

// Core 1 only; blocks its caller, never core 0's foreground loop.
static bool core0_request(uint8_t operation) {
    flash_service_request_t request = { operation };
    queue_add_blocking(&flash_service_request_queue, &request);
    bool ok = false;
    queue_remove_blocking(&flash_service_result_queue, &ok);
    return ok;
}

enum { FLASH_JOURNAL_MAGIC = 0x314C4E4Au }; // "JNL1".

typedef struct {
    uint32_t magic;
    uint32_t sequence;
    uint32_t target_offset;
    uint32_t data_crc32;
    uint32_t header_crc32;
    uint8_t reserved[FLASH_PAGE_SIZE - 20u];
} flash_journal_header_t;

_Static_assert(sizeof(flash_journal_header_t) == FLASH_PAGE_SIZE,
    "journal header must occupy one flash page");

typedef struct {
    uint32_t header_offset;
    uint32_t data_offset;
    uint32_t target_offset;
    uint8_t *data;
    flash_journal_header_t header;
    bool committed;
} flash_write_params_t;

static bool flash_disk_block_offset_valid(uint32_t offset) {
    uint32_t disk_end = FLASH_DISK_BASE_OFFSET +
        FLASH_DISK_SLOT_COUNT * FLASH_DISK_SLOT_BYTES;
    return offset >= FLASH_DISK_BASE_OFFSET &&
        offset <= disk_end - FLASH_SECTOR_SIZE &&
        (offset & (FLASH_SECTOR_SIZE - 1u)) == 0;
}

static _Noreturn void reboot_after_flash_safe_failure(void) {
    gpio_put(PIN_RESET_N, 0);
    isolate_buses();
    stop_z80_clock();
    watchdog_reboot(0, 0, 0);
    while (true)
        tight_loop_contents();
}

static void flash_write_callback(void *param) {
    flash_write_params_t *p = (flash_write_params_t *)param;
    flash_range_erase(p->header_offset, FLASH_JOURNAL_PAIR_BYTES);
    flash_range_program(p->data_offset, p->data, FLASH_SECTOR_SIZE);
    flash_range_program(p->header_offset, (const uint8_t *)&p->header,
        FLASH_PAGE_SIZE);
    flash_range_erase(p->target_offset, FLASH_SECTOR_SIZE);
    flash_range_program(p->target_offset, p->data, FLASH_SECTOR_SIZE);

    p->committed = memcmp((const void *) (XIP_BASE + p->target_offset),
        p->data, FLASH_SECTOR_SIZE) == 0;
    if (p->committed) {
        flash_range_erase(p->header_offset, FLASH_SECTOR_SIZE);
        p->committed = *(const uint32_t *) (XIP_BASE + p->header_offset) ==
            UINT32_MAX;
    }
}

typedef struct {
    uint32_t header_offset;
    uint32_t target_offset;
    uint8_t *data;
    bool restored;
} flash_restore_params_t;

static void flash_restore_callback(void *param) {
    flash_restore_params_t *p = (flash_restore_params_t *)param;
    flash_range_erase(p->target_offset, FLASH_SECTOR_SIZE);
    flash_range_program(p->target_offset, p->data, FLASH_SECTOR_SIZE);
    p->restored = memcmp((const void *) (XIP_BASE + p->target_offset),
        p->data, FLASH_SECTOR_SIZE) == 0;
    if (p->restored) {

```

```

    flash_range_erase(p->header_offset, FLASH_SECTOR_SIZE);
    p->restored = *(const uint32_t *) (XIP_BASE + p->header_offset) ==
        UINT32_MAX;
}
}

// Core 0 only, before core 1 launch and before RESET# release.
static bool flash_recover_journal(void) {
    static flash_journal_header_t headers[FLASH_JOURNAL_PAIR_COUNT];
    static bool valid[FLASH_JOURNAL_PAIR_COUNT];
    static uint8_t recovery_block[FLASH_SECTOR_SIZE];

    for (unsigned pair = 0; pair < FLASH_JOURNAL_PAIR_COUNT; ++pair) {
        uint32_t header_offset = FLASH_JOURNAL_BASE_OFFSET +
            pair * FLASH_JOURNAL_PAIR_BYTES;
        memcpy(&headers[pair], (const void *) (XIP_BASE + header_offset),
            sizeof headers[pair]);
        valid[pair] = headers[pair].magic == FLASH_JOURNAL_MAGIC &&
            headers[pair].header_crc32 == crc32_bytes((const uint8_t *)&headers[pair],
                offsetof(flash_journal_header_t, header_crc32)) &&
            flash_disk_block_offset_valid(headers[pair].target_offset);
    }

    for (unsigned recovered = 0; recovered < FLASH_JOURNAL_PAIR_COUNT;
        ++recovered) {
        int selected = -1;
        for (unsigned pair = 0; pair < FLASH_JOURNAL_PAIR_COUNT; ++pair) {
            if (valid[pair] && (selected < 0 || headers[pair].sequence <
                headers[(unsigned)selected].sequence))
                selected = (int)pair;
        }
        if (selected < 0)
            break;

        unsigned pair = (unsigned)selected;
        uint32_t header_offset = FLASH_JOURNAL_BASE_OFFSET +
            pair * FLASH_JOURNAL_PAIR_BYTES;
        uint32_t data_offset = header_offset + FLASH_SECTOR_SIZE;
        memcpy(recovery_block, (const void *) (XIP_BASE + data_offset),
            sizeof recovery_block);
        if (crc32_bytes(recovery_block, sizeof recovery_block) !=
            headers[pair].data_crc32)
            return false;

        flash_restore_params_t params = {
            header_offset, headers[pair].target_offset, recovery_block, false
        };
        int rc = flash_safe_execute(flash_restore_callback, &params, 1000);
        if (rc != PICO_OK || !params.restored)
            return false;
        valid[pair] = true;
    }
    return true;
}

// Core 1 only. Rewrites one flash sector of a CP/M disk slot, freezing
// the Z80 for the whole operation so its PWM-driven clock never
// advances while flash_safe_execute() pauses XIP fetches on both cores
// (Section 6.3). Core 0 must call flash_safe_execute_core_init() before
// core 1 starts, because core 0 is the victim parked by this call.
static bool flash_disk_write_sector(unsigned drive, uint32_t sector_offset,
    const uint8_t *data, size_t length) {
    uint32_t within_block = sector_offset & (FLASH_SECTOR_SIZE - 1u);
    if (drive >= FLASH_DISK_SLOT_COUNT || data == NULL || length == 0 ||
        sector_offset >= FLASH_DISK_SLOT_BYTES ||
        length > FLASH_DISK_SLOT_BYTES - sector_offset ||
        length > FLASH_SECTOR_SIZE - within_block)
        return false;

    uint32_t block_offset = sector_offset - within_block;
    uint32_t flash_offset = FLASH_DISK_BASE_OFFSET +
        drive * FLASH_DISK_SLOT_BYTES + block_offset;

    static uint8_t block[FLASH_SECTOR_SIZE];
    memcpy(block, (const uint8_t *) (XIP_BASE + flash_offset), sizeof block);
    memcpy(block + within_block, data, length);

    if (!core0_request(FLASH_SERVICE_ACQUIRE_BUS))
        return false;

    static uint32_t sequence;
    static unsigned next_pair;

```



```

unsigned pair = next_pair++ % FLASH_JOURNAL_PAIR_COUNT;
uint32_t header_offset = FLASH_JOURNAL_BASE_OFFSET +
    pair * FLASH_JOURNAL_PAIR_BYTES;

flash_write_params_t params;
memset(&params, 0, sizeof params);
params.header_offset = header_offset;
params.data_offset = header_offset + FLASH_SECTOR_SIZE;
params.target_offset = flash_offset;
params.data = block;
memset(&params.header, 0xFF, sizeof params.header);
params.header.magic = FLASH_JOURNAL_MAGIC;
params.header.sequence = ++sequence;
params.header.target_offset = flash_offset;
params.header.data_crc32 = crc32_bytes(block, sizeof block);
params.header.header_crc32 = crc32_bytes((const uint8_t *)&params.header,
    offsetof(flash_journal_header_t, header_crc32));

int rc = flash_safe_execute(flash_write_callback, &params, 1000);
if (rc != PICO_OK)
    reboot_after_flash_safe_failure();
bool released = core0_request(FLASH_SERVICE_RELEASE_BUS);

return released && params.committed;
}

enum {
    DISK_COMMAND_STATUS_PORT = 0x10,
    DISK_DRIVE_PORT = 0x11,
    DISK_LBA_LOW_PORT = 0x12,
    DISK_LBA_HIGH_PORT = 0x13,
    DISK_DATA_PORT = 0x14,
    DISK_COMMAND_CLEAR = 0,
    DISK_COMMAND_READ = 1,
    DISK_COMMAND_WRITE = 2,
    DISK_STATUS_READY = 1u << 0,
    DISK_STATUS_DATA_READY = 1u << 1,
    DISK_STATUS_DATA_ROOM = 1u << 2,
    DISK_STATUS_BUSY = 1u << 3,
    DISK_STATUS_ERROR = 1u << 7,
    DISK_WRITE_QUEUE_DEPTH = 2
};

typedef struct {
    uint8_t drive;
    uint16_t lba;
    uint8_t data[FLASH_DISK_RECORD_BYTES];
} disk_write_request_t;

static queue_t disk_write_queue; // Z80/Core 0 -> Core 1.
static uint32_t disk_status = DISK_STATUS_READY;
static uint32_t disk_fatal_error;
static uint8_t disk_drive;
static uint16_t disk_lba;
static uint8_t disk_write_drive; // Snapshot of drive/lba at WRITE issue,
static uint16_t disk_write_lba; // immune to changes during data transfer.
static uint16_t disk_data_index;
static uint8_t disk_data[FLASH_DISK_RECORD_BYTES];

static uint32_t disk_status_load(void) {
    return __atomic_load_n(&disk_status, __ATOMIC_ACQUIRE);
}

static void disk_status_store(uint32_t status) {
    __atomic_store_n(&disk_status, status, __ATOMIC_RELEASE);
}

static bool disk_address_valid(void) {
    return disk_drive < FLASH_DISK_SLOT_COUNT &&
        disk_lba < FLASH_DISK_RECORD_COUNT;
}

static void disk_service_init(void) {
    queue_init(&disk_write_queue, sizeof(disk_write_request_t),
        DISK_WRITE_QUEUE_DEPTH);
    __atomic_store_n(&disk_fatal_error, 0, __ATOMIC_RELEASE);
    disk_status_store(DISK_STATUS_READY);
}

static void disk_start_command(uint8_t command) {
    if (disk_status_load() & DISK_STATUS_BUSY)
        return;

```

```

bool fatal = __atomic_load_n(&disk_fatal_error, __ATOMIC_ACQUIRE);
if (command == DISK_COMMAND_CLEAR) {
    disk_data_index = 0;
    disk_status_store(DISK_STATUS_READY | (fatal ? DISK_STATUS_ERROR : 0));
    return;
}
if (fatal) {
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
    return;
}
if (!disk_address_valid()) {
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
    return;
}

disk_data_index = 0;
if (command == DISK_COMMAND_READ) {
    const uint8_t *source = flash_disk_slot_ptr(disk_drive) +
        (uint32_t)disk_lba * FLASH_DISK_RECORD_BYTES;
    memcpy(disk_data, source, sizeof disk_data);
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_DATA_READY);
} else if (command == DISK_COMMAND_WRITE) {
    disk_write_drive = disk_drive;
    disk_write_lba = disk_lba;
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_DATA_ROOM);
} else {
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
}
}

static uint8_t disk_virtual_io_read(uint8_t port) {
    if (port == DISK_COMMAND_STATUS_PORT)
        return (uint8_t)disk_status_load();
    if (port != DISK_DATA_PORT ||
        !(disk_status_load() & DISK_STATUS_DATA_READY))
        return 0;

    uint8_t value = disk_data[disk_data_index++];
    if (disk_data_index == sizeof disk_data)
        disk_status_store(DISK_STATUS_READY);
    return value;
}

static void disk_virtual_io_write(uint8_t port, uint8_t value) {
    uint32_t status = disk_status_load();
    if (port == DISK_COMMAND_STATUS_PORT) {
        disk_start_command(value);
    } else if (status & DISK_STATUS_BUSY) {
        return;
    } else if (port == DISK_DRIVE_PORT) {
        disk_drive = value;
    } else if (port == DISK_LBA_LOW_PORT) {
        disk_lba = (uint16_t)((disk_lba & 0xFF00u) | value);
    } else if (port == DISK_LBA_HIGH_PORT) {
        disk_lba = (uint16_t)((disk_lba & 0x00FFu) | ((uint16_t)value << 8));
    } else if (port == DISK_DATA_PORT &&
        (status & DISK_STATUS_DATA_ROOM)) {
        disk_data[disk_data_index++] = value;
        if (disk_data_index == sizeof disk_data) {
            disk_write_request_t request = {
                .drive = disk_write_drive, .lba = disk_write_lba
            };
            memcpy(request.data, disk_data, sizeof request.data);
            if (queue_try_add(&disk_write_queue, &request))
                disk_status_store(DISK_STATUS_BUSY);
            else
                disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
        }
    }
}

// Core 1 only; call on every foreground-loop iteration, including while
// Wi-Fi is disconnected.
static void core1_service_disk_write(void) {
    disk_write_request_t request;
    if (!queue_try_remove(&disk_write_queue, &request))
        return;

    uint32_t offset = (uint32_t)request.lba * FLASH_DISK_RECORD_BYTES;
    bool ok = flash_disk_write_sector(request.drive, offset,
        request.data, sizeof request.data);

```

```

if (!ok)
    __atomic_store_n(&disk_fatal_error, 1, __ATOMIC_RELEASE);
disk_status_store(DISK_STATUS_READY | (ok ? 0 : DISK_STATUS_ERROR));
}

```

The Z80 BIOS sets drive and 16-bit LBA, then writes command 1 for a read or command 2 for a write. A read returns 128 bytes from the data port; a write accepts exactly 128 bytes there, then changes status to BUSY until core 1 finishes the journaled update. Command 0 clears a transient protocol/queue error when not busy; a flash or journal failure remains latched until reboot and recovery. The BIOS must poll READY/DATA_READY/DATA_ROOM/BUSY instead of assuming host timing. No erase, program, blocking queue, or flash-safe call runs inside `io_trap_handler()`.

WebSocket Terminal I/O Bridge (Final Phase 10 Integration)

Maintained source: [terminal.h](#), [terminal_bridge.c](#), and [terminal_network.cpp](#).

The terminal bridge follows the `pico-altair-8800` model: initialize the queues on core 0, load the boot image from flash (Section 6.3, entirely on core 0), then launch core 1 to own Wi-Fi and WebSocket work -- and, per Section 6.3, the flash disk-write service, in the same task. The exact HTTP/WebSocket library can be `pico-ws-server` as in that project, or another lwIP-based server with the same callback shape. Only the queue functions are visible to the Z80 trap.

```

#include "pico/time.h"
#include "pico/multicore.h"
#include "pico/util/queue.h"

enum {
    TERM_DATA_PORT = 0x00,
    TERM_STATUS_PORT = 0x01,
    TERM_RX_DEPTH = 128,
    TERM_TX_DEPTH = 512,
    TERM_STATUS_RX_READY = 1u << 0,
    TERM_STATUS_TX_ROOM = 1u << 1,
    TERM_STATUS_CLIENT = 1u << 7
};

static queue_t terminal_rx_queue; // Browser/Core 1 -> Z80/Core 0.
static queue_t terminal_tx_queue; // Z80/Core 0 -> Browser/Core 1.
static uint32_t terminal_client_connected;
static uint32_t terminal_rx_drop_count;
static uint32_t terminal_tx_drop_count;

static void terminal_queues_init(void) {
    queue_init(&terminal_rx_queue, sizeof(uint8_t), TERM_RX_DEPTH);
    queue_init(&terminal_tx_queue, sizeof(uint8_t), TERM_TX_DEPTH);
}

uint8_t process_virtual_io_read(uint8_t port) {
    if (port >= DISK_COMMAND_STATUS_PORT && port <= DISK_DATA_PORT)
        return disk_virtual_io_read(port);

    if (port == TERM_DATA_PORT) {
        uint8_t value = 0;
        queue_try_remove(&terminal_rx_queue, &value);
        return value;
    }

    if (port == TERM_STATUS_PORT) {
        uint8_t status = 0;
        if (queue_get_level(&terminal_rx_queue) != 0)
            status |= TERM_STATUS_RX_READY;
        if (queue_get_level(&terminal_tx_queue) < TERM_TX_DEPTH)
            status |= TERM_STATUS_TX_ROOM;
        if (__atomic_load_n(&terminal_client_connected, __ATOMIC_ACQUIRE))
            status |= TERM_STATUS_CLIENT;
        return status;
    }

    return 0xFF;
}

void process_virtual_io_write(uint8_t port, uint8_t value) {
    if (port >= DISK_COMMAND_STATUS_PORT && port <= DISK_DATA_PORT) {
        disk_virtual_io_write(port, value);
        return;
    }
}

```

```

if (port != TERM_DATA_PORT)
    return;
if (!queue_try_add(&terminal_tx_queue, &value))
    __atomic_fetch_add(&terminal_tx_drop_count, 1, __ATOMIC_RELAXED);
}

// Called by the WebSocket server on core 1 when browser bytes arrive.
static bool terminal_ws_receive(const uint8_t *payload, size_t length,
    void *user_data) {
    (void)user_data;
    for (size_t i = 0; i < length; ++i) {
        uint8_t value = payload[i] == '\n' ? '\r' : payload[i];
        if (!queue_try_add(&terminal_rx_queue, &value)) {
            uint8_t discard;
            queue_try_remove(&terminal_rx_queue, &discard);
            if (!queue_try_add(&terminal_rx_queue, &value))
                __atomic_fetch_add(&terminal_rx_drop_count, 1, __ATOMIC_RELAXED);
        }
    }
    return true;
}

// Called by the WebSocket server on core 1 when it can send browser data.
static size_t terminal_ws_supply(uint8_t *buffer, size_t max_length,
    void *user_data) {
    (void)user_data;
    size_t count = 0;
    while (count < max_length && queue_try_remove(&terminal_tx_queue,
        &buffer[count]))
        ++count;
    return count;
}

static void terminal_ws_connected(void *user_data) {
    (void)user_data;
    __atomic_store_n(&terminal_client_connected, 1, __ATOMIC_RELEASE);
}

static void terminal_ws_disconnected(void *user_data) {
    (void)user_data;
    __atomic_store_n(&terminal_client_connected, 0, __ATOMIC_RELEASE);
    uint8_t discard;
    while (queue_try_remove(&terminal_rx_queue, &discard)) {}
    while (queue_try_remove(&terminal_tx_queue, &discard)) {}
}

enum { WS_OUTPUT_TIMER_INTERVAL_MS = 20, WS_INPUT_TIMER_INTERVAL_MS = 10 };

static uint32_t pending_ws_output;
static uint32_t pending_ws_input;
static struct repeating_timer ws_output_timer;
static struct repeating_timer ws_input_timer;

static bool ws_output_timer_callback(struct repeating_timer *t) {
    (void)t;
    __atomic_store_n(&pending_ws_output, 1, __ATOMIC_RELEASE);
    return true; // Keep repeating.
}

static bool ws_input_timer_callback(struct repeating_timer *t) {
    (void)t;
    __atomic_store_n(&pending_ws_input, 1, __ATOMIC_RELEASE);
    return true;
}

bool wifi_service_poll(void);
void terminal_websocket_server_start(uint16_t port,
    bool (*receive)(const uint8_t *, size_t, void *),
    size_t (*supply)(uint8_t *, size_t, void *),
    void (*connected)(void *), void (*disconnected)(void *));
void terminal_websocket_server_poll_output(void);
void terminal_websocket_server_poll_input(void);
void supervisor_usb_poll_nonblocking(void);

// The single core 1 entry point: WebSocket terminal and the Section 6.3
// flash disk-write service share this one task, as
// 'multicore_launch_core1()' only accepts one function. The boot image
// is already in SRAM by the time this runs (Section 6.3/8.10).
static void core1_main(void) {
    bool websocket_started = false;

    while (true) {

```

```

core1_service_disk_write();

bool network_ready = wifi_service_poll();
if (network_ready && !websocket_started) {
    terminal_websocket_server_start(8088, terminal_ws_receive,
    terminal_ws_supply, terminal_ws_connected, terminal_ws_disconnected);
    websocket_started = true;
}
if (network_ready && websocket_started &&
    __atomic_exchange_n(&pending_ws_output, 0, __ATOMIC_ACQ_REL)) {
    terminal_websocket_server_poll_output();
}
if (network_ready && websocket_started &&
    __atomic_exchange_n(&pending_ws_input, 0, __ATOMIC_ACQ_REL)) {
    terminal_websocket_server_poll_input();
}
tight_loop_contents();
}
}

static bool start_core1_services(void) {
    terminal_queues_init();           // Core 0 creates queues before launch.
    flash_service_queues_init();
    disk_service_init();
    if (!flash_safe_execute_core_init())
        return false;               // Core 0 registers as lockout victim.
    if (!add_repeating_timer_ms(-WS_OUTPUT_TIMER_INTERVAL_MS,
    ws_output_timer_callback, NULL, &ws_output_timer))
        return false;
    if (!add_repeating_timer_ms(-WS_INPUT_TIMER_INTERVAL_MS,
    ws_input_timer_callback, NULL, &ws_input_timer)) {
        cancel_repeating_timer(&ws_output_timer);
        return false;
    }
    multicore_launch_core1(core1_main);
    return true;
}

static void supervisor_fail_closed(const char *reason) {
    gpio_put(PIN_RESET_N, 0);
    isolate_buses();
    stop_z80_clock();
    printf("supervisor halted: %s\n", reason);
    while (true)
        tight_loop_contents();
}

int main(void) {
    diagnostic_safe_startup();        // First GPIO action; RESET# stays LOW.
    stdio_init_all();
    mcp_spi_init();

    if (!boot_cpm_from_flash())
        supervisor_fail_closed("boot package or journal recovery failed");
    if (!start_core1_services())
        supervisor_fail_closed("flash lockout or timer initialization failed");
    if (!set_z80_clock_hz(1000000))
        supervisor_fail_closed("invalid Z80 clock configuration");

    enable_io_trap();                // Arm before the Z80 can issue I/O.
    gpio_put(PIN_RESET_N, 1);        // Boot image verified; begin execution.

    while (true) {
        core0_service_flash_requests();
        supervisor_usb_poll_nonblocking();
        tight_loop_contents();
    }
}

```

wifi_service_poll(), terminal_websocket_server_start(), and the two server poll functions stand for the network layer, not new Z80-facing logic. Their implementation belongs entirely to core 1 and should mirror the reference project's core1_io_mgr.c pattern. wifi_service_poll() is an idempotent, bounded lifecycle state machine: it initializes CYW43, enables station mode, and associates without sleeping; after a partial initialization failure it cleans up and retries with an internal backoff. Core 1 calls it on every loop even after the server starts. It returns false while unavailable, re-enters association after link loss, and returns true once the station link is usable again. This guarantees core1_service_disk_write() runs even with Wi-Fi absent or reconnecting. Once associated, disable power-saving with

cyw43_wifi_pm(&cyw43_state, CYW43_NO_POWERSAVE_MODE) for lower terminal latency.
supervisor_usb_poll_nonblocking() similarly stands for an optional command parser that must return promptly so core 0 cannot starve flash ownership requests.

Required Integration Order

Maintained source: [Stage 10 main.c](#) and [Stage 10 CMakeLists.txt](#).

The command-loop application must call `diagnostic_safe_startup()` as its first GPIO action. After Phase 3 hardware is fitted, call `mcp_spi_init()` before any MCP access. Keep `enable_io_trap()` disabled during DMA and single-step operation; call `set_z80_clock_hz()` first, then enable the trap immediately before releasing RESET# for a PWM-run test. Before returning to DMA from a running CPU, call `disable_io_trap()` and either request the CPU bus while the clock still runs or assert RESET# and supply at least three further full clocks. Only after the selected ownership procedure completes and the Z80 bus is verified high-impedance may the firmware enable either transceiver. While the trap is enabled, reserve SPI0 for the handler: no other IRQ, core, or main-loop operation may access the MCP23S17. Drain `core0_service_flash_requests()` continuously from core 0's nonblocking foreground loop. For acquisition, leave trapping enabled while asserting BUSREQ# and waiting for BUSACK# LOW, then disable it. For release, enable trapping while BUSACK# is still LOW, then deassert BUSREQ# and wait for BUSACK# HIGH. Core 0 must call and check `flash_safe_execute_core_init()` before launching core 1 and must not use the multicore FIFO for anything else, because the lockout handler owns it. Journal recovery is the only core-0 flash write and occurs before core 1 launch; all runtime writes execute on core 1 while the Z80 is held in BUSACK#.

8.14 Datasheet References

- [Z84C00 CPU datasheet](#)
- [AS6C1008 SRAM datasheet](#)
- [MCP23017/MCP23S17 datasheet](#)
- [ATF22V10C datasheet](#)
- [SN74HCT541 datasheet](#)
- [SN74AHCT245 datasheet](#)
- [SN74LVC245A datasheet](#)
- [SN74LVC244A datasheet](#)
- [1N5817/1N5818/1N5819 Schottky diode datasheet](#)
- [RP2350 datasheet](#)
- [Raspberry Pi Pico 2 board datasheet](#)
- [BusBoard BB830 breadboard datasheet](#)

8.15 Source Code Index

Canonical repository: github.com/gloveboxes/Z80ROMlessSBC. All phase applications are cumulative: each stage links the shared modules proven by earlier stages, while remaining independently buildable for hardware bring-up.

Project Build Files

File	Purpose
CMakeLists.txt	Pico SDK import, Pico 2 W target, project languages, and protected firmware flash linker boundary
src/CMakeLists.txt	Shared firmware libraries, stage registration, and the z80_cpm_images artifact target
.gitignore	Generated build, Python cache, and assembler-output exclusions
package.json, package-lock.json	Locked Mermaid and browser-automation dependencies plus the npm run pdf command
scripts/build-readme-pdf.mjs	Pandoc, Mermaid, and Chromium PDF renderer with full-page diagram layout

Cumulative Stage Applications

Phase	Application	Target definition	Responsibility
0	Power checklist	Hardware-only	Empty-socket wiring, rail, resistance, and startup-state checks
1	main.c	CMakeLists.txt	Safe supervisor startup and GPIO walk
2	main.c	CMakeLists.txt	Buffered controls and variable-frequency clock
3	main.c	CMakeLists.txt	MCP23S17 register and port diagnostics
4	main.c	CMakeLists.txt	16-bit address-bus drive, sample, and isolation tests
5	main.c	CMakeLists.txt	8-bit data-bus drive, sample, and isolation tests
6	main.c	CMakeLists.txt	SRAM DMA and full-memory validation
7	main.c	CMakeLists.txt	Z80 reset, execution, clock, and BUSREQ/BUSACK tests
8	main.c	CMakeLists.txt	Virtual-ROM preload and synchronous I/O trapping
9	main.c	CMakeLists.txt	Manifest boot, journal recovery, and persistent flash disks
10	main.c	CMakeLists.txt	Final CP/M, flash-disk, Wi-Fi, and WebSocket integration

Shared Firmware Modules

Module	Public interface	Implementation	Responsibility
Pin map	pins.h	Header-only	Authoritative Pico GPIO assignments
Supervisor	supervisor.h	supervisor.c	Safe GPIO startup and fail-safe isolation defaults
Clock	clock.h	clock.c	PWM frequency selection, stop/resume, and single-cycle clocking
MCP23S17	mcp23s17.h	mcp23s17.c	SPI register access and 16-bit address expansion
Buses	bus.h	bus.c	Contention-safe address/data direction, isolation, drive, and sample operations
SRAM	sram.h	sram.c	DMA byte access, image load/verify, and RAM diagnostics
CPU	cpu.h	cpu.c	Reset-held DMA, run control, bus ownership, and fail-closed state
I/O trap	io_trap.h	io_trap.c	Synchronous Z80 IN/OUT interception and fault counters
Flash disk	flash_disk.h	disk_device.c	Z80-facing command/status, drive, LBA, and 128-byte data ports
Flash layout	flash_layout.h	Header-only	Firmware, journal, boot-package, and disk-slot offsets
Flash backend	flash_backend.h	flash_backend.c	Manifest validation, SRAM boot load, journal recovery, and core-1 commits
Terminal	terminal.h	terminal_bridge.c, terminal_network.cpp	Queue-backed terminal ports, Wi-Fi lifecycle, HTTP, and WebSocket service

CP/M, Disk, and Web Tooling

Area	Files	Purpose
Native CP/M	README, bios.asm, build_images.py, test_build_images.py	Native BIOS, CCP/BDOS extraction, boot package, full-flash composition, and host regression tests
Disk geometry and conversion	README, geometry.py, convert_altair_disks.py	Shared CP/M geometry and deterministic Altair-media conversion
Generated disk artifacts	generated directory, manifest.json	Four exact 320 KiB intermediate disk slots and source/output hashes
Preserved source media	source-altair directory, altair_88diskrom.h, altair_disk_loader.h	Original framed disks, Altair loader references, and upstream license
Browser terminal	terminal.html, embed_html.py	Embedded Stage 10 terminal client and build-time HTML conversion
Network configuration	lwipopts.h, wifi_config.h.in	lwIP settings and build-time Wi-Fi credential template

Generate the PDF

The PDF renderer requires Node.js, Pandoc, and Microsoft Edge, Google Chrome, or Chromium. On macOS, install the command-line prerequisites with:

```
brew install node pandoc
```

From the repository root, install the locked dependencies and generate README.pdf:

```
npm ci  
npm run pdf
```

The renderer preserves tables, code highlighting, local images, links, and MathML. Mermaid diagrams are emitted as vector graphics on dedicated pages; wide diagrams automatically use A4 landscape orientation.

Set BROWSER_EXECUTABLE if the browser is installed in a nonstandard location:

```
BROWSER_EXECUTABLE="/path/to/chrome" npm run pdf
```

To render another Markdown file or choose another output path, pass both paths after `--`:

```
npm run pdf -- README.md README.pdf
```